Insurers Monitor Shocks to Collateral: Micro Evidence from Mortgage-backed Securities*

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Abstract

This paper studies how insurance companies respond to anticipated cash flow shocks in commercial mortgage-backed securities (CMBS). Using granular micro data around the onset of the COVID-19 pandemic, we show that upcoming lease expirations predict delinquency on commercial real estate mortgages, and that this relationship strengthens for office properties after the pandemic-induced decline in office demand. Insurers subsequently sell CMBS with greater exposure to expiring office leases, particularly non-senior tranches, where information about collateral risk matters most. Our findings indicate that insurance companies actively monitor the collateral underlying complex securitized assets, going beyond regulatory and market signals.

JEL codes: G20, G21, G22, G23

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1. Introduction

Growing risks in mortgage-backed securities, along with perceived failure by intermediaries to perform due diligence and risk management, are considered some of the main causes of the Global Financial Crisis (Chen et al., 2020). For commercial mortgage-backed securities (CMBS), risk assessment is particularly challenging because cash flows depend on the performance of multiple underlying mortgages and are redistributed through highly complex securitization structures (Ghent, Torous and Valkanov, 2019). This complexity limits transparency and can obscure economically important characteristics of the underlying loans. Consistent with this view, Griffin and Priest (2023) show that investors are not compensated for exposure to risks arising from income overstatement at origination. It is therefore crucial to understand if investors react to cash flow risks in CMBS, as investor reactions to salient cash flow shocks can reveal whether and how they monitor risks in these complex securities.

In this paper, we examine how cash flow shocks driven by lease expirations and structural changes in office demand affect both mortgage performance and institutional investor behavior. Using detailed micro data around the onset of the COVID-19 pandemic, we show that lease expiration predicts commercial real estate mortgage delinquency, and particularly so for offices. Insurers—one of the largest investor groups in mortgage-backed securities—sell more exposed CMBS before delinquencies materialize, suggesting they actively monitor these risks. Moreover, this effect is more pronounced for non-senior tranches, for which information about collateral risk is more consequential.

To understand the extent to which these trading responses reflect monitoring by insurers, we first establish whether lease expirations actually predict delinquency in a way that sophisticated investors could anticipate. Using comprehensive data on commercial mortgages in CMBS deals, containing detailed loan and property characteristics as well as information about lease contracts between borrowers and their core tenants, we find that lease expiration increases CRE loan default risk for offices, especially following COVID-19 when hybrid work arrangements reduced office demand (Barrero, Bloom and Davis, 2021) and, thus, rental income from CRE.

Using the COVID-19 pandemic as a shock to the demand for office space, we estimate a difference-in-differences specification and show that lease expiration triggers increases in

delinquencies, with a stronger effect after COVID-19. These effects are economically meaningful, with lease expiration being associated with about 1.3 percentage points higher delinquency in the post-pandemic period, and an additional 1.2 percentage points increase among office-linked loans.

A challenge in establishing a causal link between lease expiration dates and delinquency is that these dates could coincide with other shocks that cause delinquency, e.g., other shocks that lower demand for CRE. Similarly, if mortgages with leases expiring have floating interest rates, concurrent increases in reference interest rates can also cause an increase in delinquency rates. We address these challenges by leveraging the granularity of our data, which allow us to include a rich set of fixed effects that capture several static and time-varying confounding factors that could affect delinquency. In addition, our identification strategy exploits lease expiration dates set well before the COVID-19 shock, so that the timing of lease termination is plausibly exogenous to pandemic-induced structural shifts in CRE demand.

We then document how large insurance companies' exposure to offices through their CMBS holdings is and the extent to which these investors monitor underlying CMBS collateral characteristics, including cash flow risk caused by lease expirations. If insurers monitor lease-level information and anticipate worse lease-rollover risk for offices after the pandemic, then they should be more likely to sell CMBS with a larger share of mortgages linked to leases expiring when faced with unexpected shocks to collateral demand.

To evaluate this possibility, we combine our comprehensive monthly panel data on CMBS deals and mortgages against CRE with granular information on the asset portfolios of U.S. insurance companies. Insurance companies are indeed a large group of investors in CMBS, holding close to one-fifth of newly issued private-label CMBS between 2017 and 2022. We also find that the fraction of insurers' private-label CMBS portfolio exposed to offices peaks in 2020, and decreases afterwards, which is consistent with lower demand for CMBS with office exposures among those investors.

We first assess whether insurers seem to monitor time-varying characteristics of underlying CMBS collateral. We do so by testing if insurers are more likely to sell tranches that are linked to watchlisted mortgages, mortgages whose properties have lower or declining occupancy rates, or tranches with declining financial performance metrics such as net operating income and debt service coverage ratios. In all cases, we find that insurers are more likely to

sell tranches whose characteristics indicate deteriorating collateral performance. Moreover, the richness of our data allows us to include insurer by time and insurer by bond fixed effects, in addition to coupon type, risk classification, and lagged downgrade by year fixed effects. This addresses concerns that our estimates may be contaminated by other time-varying insurer shocks or bond characteristics. It also allows us to capture changes in trading behavior within an asset class with similar capital costs for insurers, and rules out portfolio rebalancing in response to ratings downgrades.

These correlations suggest that insurers react to a broad set of granular collateral-performance signals. Since these variables reflect realized performance and are potentially endogenous, we turn to studying the effect of the unexpected COVID-19 shock on lease-rollover risk and its impact on mortgage cash flows. In particular, we test if insurers monitor cash flow risks in their CMBS portfolio by asking if bonds more sensitive to different cash flow shocks are more likely to be sold following the sudden, unexpected increase in risk caused by COVID-19. Our identification strategy relies on the idea that pandemic-driven lower demand for CRE constitutes an unexpected shock to CMBS cash flows, with different effects across property collateral types. We find that, for the average CMBS in our sample, the majority of their cash flow shocks induced by lease expiration occurs within six years. Informed by this threshold, we estimate a difference-in-differences specification to assess whether CMBS with exposure to office-linked loans whose main leases expire within six years are more likely to be sold after the pandemic.

Consistent with the idea that they infer risks from shocks to expected cash flows, we find that insurance companies are more likely to sell CMBS that are exposed to offices with leases expiring within six years. This result is also robust to accounting for other CMBS characteristics that correlate with our treatment definition. This suggests that they can identify how different property types are affected by the pandemic and the nature of cash flow risks caused by lower demand for offices. Treated bonds (those exposed to office lease expiration within six years) are over 2 percentage points more likely to be sold by insurers in the post-COVID period than comparable bonds that are not exposed to office lease expiration within six years.

We further corroborate the importance of the increase in lease-rollover risk by defining a measure of the present discounted value of lease-expiration exposures. This measure uses the complete lease-expiration schedule across all major tenants for each property that is linked to

a mortgage serving as collateral for each CMBS. Using this measure in a similar difference-in-differences specification, we find that insurers are more likely to sell bonds that have a relatively larger present discounted value of lease-expiration exposures. Moreover, the likelihood of sales increases monotonically with the size of the exposure.

The tranching structure of mortgage-backed securities implies that non-senior bonds and bonds with a smaller subordination fraction should be relatively more informationally sensitive than senior bonds and those with a higher subordination percentage, since the former tranches are the first to absorb potential losses. Hence, non-senior CMBS tranches should be subject to more intensive monitoring of expected cash flow shocks. Indeed, we find that among bonds exposed to cash flow shocks, non-senior bonds are approximately 3 to 5 percentage points more likely to be sold by insurers after the pandemic. A similar pattern arises for bond subordination, with treated tranches with a lower subordination percentage being more likely to be sold by insurers post-COVID. This is in line with the idea that monitoring of an asset's underlying cash flow risk should be more valuable when that risk is more consequential to investors.

We further distinguish this anticipatory trading from sales following observable deterioration or regulatory pressure. If insurers only dispose of exposed CMBS after adverse information has already translated into losses, their bond sales would be associated with worse realized performance. We test this mechanism by comparing realized gains and losses from the sales of treated and non-treated bonds. We find no evidence of poorer realized performance for treated bonds; if anything, insurers earn relatively higher trading returns on office-exposed CMBS. This pattern contrasts with sales following credit ratings downgrades—events that are typically associated with price declines and forced sales (Ellul, Jotikasthira and Lundblad, 2011).

We conclude our analysis by documenting several patterns of the CMBS holdings of banks. We document a striking rise in the holdings of private-label CMBS by banks after 2020, especially by small and medium-sized banks. The number of small banks that hold private-label CMBS *nearly doubles* between 2020 and 2023. Moreover, small banks' holdings of private-label CMBS also increased relative to securities assets, suggesting this rise was not merely a consequence of balance sheet expansion. Banks exposed to private-label CMBS subsequently experience weaker realized performance in their securities portfolios, with the estimated de-

cline in realized securities performance equivalent to roughly 1% of average Tier 1 capital during 2022-2023. These patterns contrast with insurers' declining exposure and illustrate how exposures to common CRE risks evolved differently across financial intermediaries after the pandemic.

Related literature. Our paper contributes to the literature studying securitized assets and mortgage-backed securities (MBS) in particular.[1] This literature emphasizes how information asymmetries and opaqueness pose challenges to investors' ability to assess and price risk, and documents how these risks affected institutional investors during the Global Financial Crisis. In the context of commercial mortgage-backed securities, Ghent, Torous and Valkanov (2019) show that more complex deals underperformed during that period, with complexity both obscuring collateral quality and enabling the diversion of cash flows toward residual tranches. Griffin and Priest (2023) document pervasive loan income overstatement at origination that persists to recent years. Importantly, neither complexity-induced risks nor income overstatement are priced in by investors, suggesting that information asymmetries and limited investor awareness remain pervasive in CMBS markets. Together, these findings underscore the difficulty of assessing risks in securitized assets, which requires costly monitoring and specialized infrastructure (Hanson and Sunderam, 2013).

Our contribution is to present evidence that despite these due-diligence challenges, institutional investors such as insurance companies monitor detailed, time-varying property and lease contract characteristics that predict CMBS losses, and they divest on the basis of such information.

As such, we also relate to a broad literature that studies insurance companies' portfolio decisions, and how they react to risks in their asset portfolio.[2] This literature documents that insurance companies react to changes in observable risk such as downgrades, and highlights how regulation affects insurers' reaction to asset risk (Ellul, Jotikasthira and Lundblad, 2011; Chen et al., 2020; Becker, Opp and Saidi, 2022; Sen, 2023). We contribute to this literature by showing that insurers adjust their CMBS portfolios in response to anticipated cash flow risks

[1] E.g., DeMarzo and Duffie (1999), DeMarzo (2005), Demiroglu and James (2012), Begley and Purnanandam (2017), Ashcraft, Gooriah and Kermani (2019), Flynn, Ghent and Tchistyi (2020), and Aiello (2022).

[2] See, among others, Chodorow-Reich, Ghent and Haddad (2021), Ge and Weisbach (2021), Koijen and Yogo (2022), Bretscher et al. (2026), Ellul et al. (2022), Koijen and Yogo (2023), O'Hara, Rapp and Zhou (2025), and Bhardwaj, Ge and Mukherjee (2025).

embedded in the collateral pool, rather than market-wide or regulatory signals. The type of monitoring we study involves assessing loan-level characteristics, such as lease rollover exposure, that predict future deterioration in performance and eventual downgrades. Unlike the fire sales documented in Ellul, Jotikasthira and Lundblad (2011), these transactions do not appear to be driven by capital constraints or regulatory pressures. This distinction highlights that insurers engage in active, information-based monitoring of complex securitized assets, rather than simply reacting to observable changes in credit quality.

In addition, we relate to the literature exploring the impact of work-from-home adjustments in CRE mortgage default risk. Thus far, this literature has not documented a direct link between lower office demand and CRE mortgage default (Van Nieuwerburgh, 2022).[3] Moreover, Jiang et al. (2023) explore how losses from CRE loan portfolios affect the solvency of U.S. banks, and Glancy and Kurtzman (2024) consider how differences in small banks' CRE loan portfolios govern reduced exposure to loans whose poor performance was driven by lower office demand. Hinzen, Severino and Nieuwerburgh (2025), however, argue that despite this lower exposure of regional banks to highly affected CRE markets, these lenders are nonetheless at risk due to under-collateralization and higher portfolio concentration.

We provide a detailed account of how insurers are affected by CRE risks through their CMBS holdings. Finally, we document that small banks substantially increased their exposure to private-label CMBS after the pandemic. We also find evidence that higher private-label CMBS holdings are associated with securities trading losses by banks after COVID-19. This highlights the importance of accounting for securities holdings, in addition to direct CRE lending, when assessing how CRE risks are distributed across financial intermediaries.

2. Lease Expiration, Cash Flow Shocks, and CRE Mortgage Default

In this section, we develop hypotheses that will guide our empirical analysis. First, sudden drops in demand for office space lead to fewer occupied offices after leases expire, either by downsizing or lack of renewal, and longer search times for new tenants. This results in

[3] As in our study, Glancy and Wang (2025) highlight the importance of lease expiration in the post-COVID period, showing that it affects office vacancies and loan performance. Both studies provide direct evidence of the importance of cash flow-triggered mortgage default for commercial real estate. Several papers study the relevance of strategic and cash-flow motives for default of *residential* mortgages (Ganong and Noel, 2023; Bhutta, Dokko and Shan, 2017; Gerardi et al., 2018), with less attention devoted to *commercial* mortgages.

lower income from new leases, reducing overall lease revenue. As a result, to the extent that borrowers rely on such income to repay mortgages, mortgage default rates should increase, especially during periods of lower demand for office space.

Hypothesis 1: *Lease expiration persistently increases defaults of mortgages against offices after the COVID-19 cash flow shock, whereas other types of collateral—especially retail—see defaults immediately and are, thus, less sensitive to lease expiration.*

Since U.S. insurers frequently hold CMBS, any increase in the riskiness of these assets could influence their investment decisions. First, if insurance companies can observe lease expirations, the increased likelihood of future delinquencies due to lease expirations should make CMBS with a higher proportion of soon-to-expire mortgages less attractive to hold. Since lower demand increases the persistence of default triggered by lease expiration, investors are more likely to monitor characteristics associated with cash flow risks following the pandemic-linked shock to CRE demand, which we infer by actions that suggest preceding acquisition and processing of related information. Hence, investors are subsequently more prone to selling CMBS with a larger exposure to cash flow shocks after the pandemic.

Hypothesis 2: *Conditional on monitoring, insurers should sell CMBS with relatively more mortgages against offices undergoing lease expiration after the COVID-19 cash flow shock.*

3. Data Description

Our data come from two main sources: Trepp and the National Association of Insurance Commissioners (NAIC). Trepp is a leading provider of commercial real estate-collateralized products data (Flynn, Ghent and Tchistyi, 2020), collecting origination information from CRE mortgages, CMBS deals and bonds, which is obtained from various sources. The data include detailed information such as property type and location, mortgage maturity, amount, interest rates, and delinquency information for each distribution date. We classify loans according to the use of the property serving as collateral for the loan. We distinguish between

Office, *Retail*, and further property types.[4] The data also contain information on lease agreements between borrowers and tenants. Our baseline results use the expiration date of the largest tenant's lease, whereas the continuous exposure measure introduced in Section 5.3 incorporates information on up to the five largest tenants of each property.[5] Appendix-Table A.1 shows that the availability of lease expiration data varies by property type, with Office and Retail as the only two property types for which the date of lease expiration of the main tenant is available for more than 50% of the observations. For this reason, we mainly consider these two property types throughout the paper.

We obtain holdings and trades of fixed income assets of all insurance companies in the U.S. from the National Association of Insurance Commissioners (NAIC). The holdings data are based on NAIC Schedule D Part 1, and contain CUSIP-level end-of-year holdings of fixed income securities, including CMBS. The trading information is obtained from NAIC Schedule D, Parts 3 and 4, which contain information on acquisitions and dispositions of fixed income assets by insurance companies, respectively. We identify actual trades (sales and purchases) using a procedure similar to that in Becker, Opp and Saidi (2022), which we describe in Appendix C.

We restrict our analysis to the post-2017 period.[6] This ensures that we mitigate concerns about the influence of the Global Financial Crisis (GFC), e.g., through elevated delinquency rates responding to demand shocks that originated during the GFC. Table 1 shows the summary statistics of the mortgages in our sample. Panel A focuses on all properties, which have a median lease expiring in 2024 and a median mortgage maturity of 10 years. We classify a loan as delinquent if payments are past due for at least 90 days. On average, less than 1% of all observations are delinquent in our sample period, around 12% of the observations have floating interest rates, and less than 1% are recourse loans.

Since our analysis mostly focuses on offices and retail CRE, we provide a breakdown of the characteristics of the mortgages used to finance these property types in Panels B and C of Table 1, respectively. Relative to retail, offices have floating interest rates more frequently,

[4] These are classified as *Multifamily*, *Mixed Use*, *Healthcare-Nursing*, *Lodging-Restaurants*, *Industrial and Warehouses*, and *Other*. The details of how these types are obtained, along with other details of our data cleaning procedure, can be found in Appendix B.

[5] Trepp only makes available information on the five largest tenants of each property.

[6] Our Trepp sample covers CMBS information until June 2022.

lower delinquency rates, and similar maturity. Moreover, the mean and the median share of each property occupied by the largest tenant is smaller in offices than in retail.

Finally, we use FDIC's Consolidated Reports of Condition and Income (Call Reports) data for our analysis of bank holdings of private-label CMBS. Further details about the data series are available in Appendix B.

4. Cash Flow Shocks and Mortgage Delinquencies

4.1. The Role of Lease Expiration

First, focusing on mortgages whose lease expiration occurs between 2017 and 2022, we evaluate the importance of lease expiration-induced cash flow shocks to borrowers in driving delinquency rates. During this sample period, we examine delinquency rates in the time window of one year prior and one year after the expiration date of the main lease. Given our definition of a loan being delinquent if it is at least 90 days, or about 3 months, past due, we expect to see delinquency rates increase comparatively more only after the third month in which a lease expires.

We explore how lease-rollover risk affects delinquency rates of different types of properties by splitting our sample into two subsamples: offices and retail properties. This allows us to assess if the impact of lease expiration on delinquency rates varies by property type. This could arise if the specific use of the property limits a borrower's ability to find a new tenant. For example, it may be more difficult to re-purpose office space for other uses, which can increase search costs and lower expected revenue after an existing lease expires. Moreover, firms in different sectors might be more likely to renew their lease contracts, and to the extent that these firms select into different types of properties, this would differentially affect borrowers depending on the property they are financing with their loan.

Figure 1 shows a remarkable difference in delinquency behavior for these different property types. The plot on the left-hand side shows a sharp increase in delinquency rates of offices four months after the end of the main lease agreement. This is in line with the idea that cash flow shocks from lease expiration induce borrowers to stop making payments on their office mortgages. This may be because the existing borrower cannot readily find a new

tenant or because the lease generates lower income than the previous one. By contrast, the plot on the right-hand side suggests that increases in delinquency rates of retail properties are more short-lived, with shocks introduced by the end of lease agreements being more transitory in nature. Overall, these results indicate that lease expiration is a strong predictor of office delinquencies, but less so for retail properties.

These differences in sensitivity of delinquency rates to lease expiration across offices and retail properties also affect the trajectory of default risk during the onset of COVID-19. Figure 2 shows the delinquency rates of properties with and without leases expiring around the start of the pandemic. We compare the average delinquency rate of loans with leases expiring in 2021-2022 with the average delinquency rate of loans without leases expiring in these two years. The left-hand side plot shows a striking pattern for office mortgages with and without leases expiring in 2021-2022. Delinquency rates for the latter group are stable throughout the entire period, whereas there is a large spike in delinquency rates of mortgages whose main leases expired in 2021-2022. This further indicates that lease expiration is a relevant determinant of office mortgage default.

By contrast, the trajectory of retail mortgage delinquencies on the right-hand side of Figure 2 shows a different pattern. Delinquency rates spike immediately at the onset of the COVID-19 pandemic, which coincides with lockdown periods during which retail stores did not generate income to tenants. Following that initial shock, mortgages with leases expiring in 2021-2022 demonstrate persistently higher delinquency rates, which suggests that lease expiration matters for the adjustment to the initial shock.

By zooming in on offices rather than retail properties, we can focus on structural changes in the demand for office space without explicitly considering the implications of the 2020 lockdowns for businesses. Furthermore, if institutional investors trade CMBS holdings before losses materialize, then one would expect their trading behavior to be based on office exposure if these mortgage losses can be predicted by lease expiration.

4.2. The Effect of Lease Expiration on Mortgage Delinquency

Our motivating evidence suggests a key role for lease expiration dates in driving delinquency behavior for CRE mortgage borrowers, especially for office properties. Nevertheless, there

are multiple other factors that could be driving the delinquency dynamics we observe for properties subject to lease expiration. For example, lease expiration dates could correlate with systematic or region-specific shocks that affect the U.S. economy in specific times, such as the Global Financial Crisis and the onset of the COVID-19 pandemic. Moreover, loans for which we have lease expiration data could also have specific characteristics, such as floating interest rates, which can make them more susceptible to increases in delinquency in times of higher interest rates.

To formally estimate the differences in post-lease expiration delinquency behavior before and after the onset of the pandemic, we estimate the following specification:

$$\begin{aligned} I_{jrt}^{D90} = {} & \alpha_j + \alpha_{rt} + \alpha_{j(floating)t} + \gamma_1 Post\ Expiration_{jt} \\ & + \beta_1 Post\ Expiration_{jt} \times\ Post\ Covid_t + \beta_2 Post\ Expiration_{jt} \times\ Ind\ Office_j \\ & + \beta_3 Post\ Covid_t \times\ Ind\ Office_j \\ & + \beta_4 Post\ Expiration_{jt} \times\ Post\ Covid_t \times\ Ind\ Office_j + \varepsilon_{jrt}, \end{aligned} \tag{1}$$

where $Post\ Covid_t$ is a dummy equal to 1 after March 2020, $Post\ Expiration_{jt}$ equals 1 if loan j had its main lease expiration before or in year-month t, and $Ind\ Office_j$ equals 1 if loan j is linked to an office. The coefficient of interest β_4 captures the difference in the effect of lease expiration-induced cash flow shocks on delinquency rates since the onset of the pandemic.

The results are in Table 2. Across all specifications, the coefficient on the triple interaction term is positive and statistically significant. The baseline effect of lease expiration on mortgage delinquency is about 1.3 to 1.4 percentage points, and increases by another 1.2 to 1.3 percentage points after the pandemic but, importantly, only for office mortgages whose main tenancy agreement expired.

In Appendix-Figure A.1, we plot the coefficients of a dynamic difference-in-differences using I_{jrt}^{D90} as the dependent variable, which is estimated splitting the sample in before and after COVID-19, and includes only office mortgages. Before the pandemic, the initial increase in delinquency falls after roughly ten months. In contrast, the increase in delinquency rates intensifies over time after COVID-19.

These results support the notion that office demand shocks after the onset of the COVID-19

pandemic exacerbate the effects of cash flow shocks on CRE mortgage delinquency rates.

CMBS exposure to regional work-from-home characteristics. Our analysis hinges on the observation that by being relatively more affected by hybrid work arrangements, demand for office properties is also relatively more affected by the COVID-19 shock, thereby leading to more persistent cash flow shocks to rent revenue. Importantly, another dimension of heterogeneity in exposure to work-from-home adjustments refers to regional characteristics. For instance, cities like San Francisco or New York are perceived to be more affected by hybrid work arrangements than others (Gupta, Mittal and Nieuwerburgh, 2023).

While we cannot directly measure demand for office space, we can nevertheless assess how mortgages in our sample correlate with measures that reflect regional sensitivity to work-from-home arrangements. For this purpose, we use the measure of jobs that can be performed remotely by Dingel and Neiman (2020), which should broadly indicate which areas are more likely to be affected by work-from-home arrangements. We then calculate the within-sample median exposure to teleworkable jobs, and re-estimate Equation (1) separately for areas with above and below exposure to teleworkable jobs. Appendix-Table A.2 shows that there is no differential COVID-19 impact on the effect of lease expiration on the delinquency rates of office mortgages in low exposure areas (columns 1-3). On the other hand, there is a large and significant increase in the effect of lease expiration on delinquency rates of office mortgages after the pandemic in high exposure areas (columns 4-6). These results highlight the connection between lower office demand after the pandemic and the increase in lease-rollover risk experienced by office-linked mortgages.

5. Do Insurers Monitor Cash Flow Risks?

We have documented a link between expected changes in the tenancy agreement of a specific office and default rates of the mortgage linked to that property, which has implications for assets whose cash flows depend on the performance of these CRE mortgages. In particular, insurance companies' cash flows obtained from their holdings of CMBS might be compromised if the underlying mortgages become non-performing.

This raises several fundamental questions. What is the extent and dynamics of the ex-

posure of insurance companies to office CRE through their holdings of CMBS? Given the predictable nature of expected cash flow shocks to mortgage payments, do insurance companies monitor such risks and sell bonds based on cash flow shocks to mortgage CRE? Finally, is this response stronger for tranches for which cash flow risks are more consequential, such as non-senior tranches? We explore these issues next.

5.1. Insurer Holdings of WFH-sensitive CMBS

We start by leveraging our data to document the importance of insurance companies for the private-label CMBS market, and to characterize their exposure to shocks linked to office collateral. First, we consider end-of-year outstanding balances and amounts issued for all private-label CMBS in our sample, and identify which bonds are held by insurance companies at the end of each year. Figure 3 shows that insurance companies are one of the main investors in CMBS markets. By the end of 2022, insurance companies held about $200 billion out of $800 billion outstanding (Panel A). Similarly, between 2017 and 2019, insurance companies acquired around 20% of the total amount of new issues of private-label CMBS in a given year (Panel B). Interestingly, the share of new CMBS originations held by insurers drops to about 17% in 2022. This reduction in insurers' share of newly issued CMBS would be consistent with lower insurer demand, which could arise as lower office demand translates to higher mortgage default rates.

We further explore the dynamics of CMBS holdings by insurance companies and document the exposure of the latter's CMBS portfolio to office CRE collateral. We classify a bond as exposed if it has *any* mortgages financing offices within its pool of collateral. We then calculate the share of CMBS exposed to offices out of the entire portfolio of private-label CMBS held by insurance companies.

Figure 4 shows the share invested in *non-exposed* bonds for each year. The share of CMBS exposed to offices increases up until 2020, at which point this trend is reversed. In particular, insurance companies increase the share of CMBS *not exposed* to offices in 2021 and 2022 by about seven percentage points. This pattern is consistent with insurers reducing their relative exposure to office-linked CMBS after COVID-19.

Broader monitoring of CMBS characteristics. The fact that insurers are prominent investors in CMBS raises the question of whether insurers monitor *any* collateral performance characteristics of these assets. For example, insurers might be concerned about lower property occupancy rates, declining financial performance of the bonds, or rely on information conveyed by inclusion in servicer watchlists.

We test this hypothesis by estimating reduced-form regressions relating insurer sales to the following CMBS performance variables: Inclusion in Servicer Watchlist, Occupancy Rates, Debt Service Coverage Ratios (DSCR), and Net Operating Income (NOI). The results are presented in Appendix-Table A.3. We find that insurers are significantly more likely to sell bonds with underlying collateral on servicer watchlists, bonds with lower occupancy rates and DSCR, and bonds linked to properties with falling occupancy rates and declining NOI and DSCR. This suggests insurers monitor CMBS collateral information across several dimensions. Nonetheless, these performance characteristics are endogenous and reflect *realized* performance. We next exploit exposure heterogeneity across CMBS to estimate the effect of *expected* cash flow shocks on insurers' trading behavior.

5.2. CMBS Exposure to Cash Flow Shocks and Trading Behavior

Insurance companies might anticipate the effect of work-from-home (WFH) shocks on the cash flows and on the value of their CMBS, and therefore attempt to sell these bonds. Moreover, even if insurance companies do not trade CMBS based on office exposure alone, they could still anticipate how shocks to their assets caused by upcoming lease expiration could lead to losses from mortgage default.

Each CMBS can be composed of multiple mortgages whose underlying leases expire in different years. This gives rise to a dynamic structure for the share of the expected losses due to default following lease expiration. Hence, the time-sensitive nature of the expected cash flow shocks, along with the structure of CMBS, generates heterogeneity in exposure to expected losses due to office mortgage default across different bonds.

To illustrate this point, we construct the distribution of expected cash flow shocks across future lease-expiration years for all CMBS held by insurance companies. For each deal in our sample, we calculate the share of its largest underlying leases expiring in τ years. Formally,

for a given deal d, we calculate:

$$Share(\%)^{\tau}_{dt} = \sum_{j \in J^d_t} \frac{\omega^{\tau}_{jt} \times Balance_{jt}}{\sum_h \omega^{h}_{jt} \times Balance_{jt}} \times 100, \tag{2}$$

where $\omega^{\tau}_{jt} \in [0, 100]$ denotes the share of mortgage j's contract at the end of year t that expires in τ years, $Balance_{jt}$ is mortgage j's outstanding balance at the end of year t, and J^d_t is the set of underlying mortgages in deal d at the end of year t.

Figure 5 shows the resulting distribution after averaging $Share(\%)^{\tau}_{dt}$ across all years in our sample, separately for offices and retail. A large share of expected cash flow shocks materializes only within three to six years relative to each year in which we obtain outstanding balances. The average bond in our sample experiences only around 16% of its cash flow shocks within two years. Hence, it would be incorrect to measure a bond's exposure to expected cash flow shocks limiting the analysis to bonds with leases expiring in the following one or two years since, on average, a large share of the shocks is yet to materialize.

The average bond in our sample has around 50% of its expected cash flow shocks to office mortgages materializing *within six years*. This provides us with a natural threshold against which to compare individual bonds. To understand if insurance companies trade based on expected cash flow shocks, we test if they sell private-label CMBS with larger exposure to office mortgages with leases expiring in the near future more frequently after the pandemic started. We define a bond j in year t as *treated* if it has leases expiring within six years relative to year t. Hence, our control group consists of bonds whose expected cash flow shocks materialize only after most cash flow shocks have *already materialized for the average CMBS*. Intuitively, if insurers discount future cash flows (at any non-zero rate), they should value losses in the short to medium term more than any long-run losses.

We then estimate the following specification:

$$\begin{aligned} I^{sold}_{ijt} =& \alpha_{it} + \alpha_{ij} + \alpha_{j(coupon)t} + \alpha_{ij(NAIC)t} + \alpha_{ij(downgrade_{t-1})t} \\ &+ \beta_1 \times Treat^{Exp\ Office}_{jt} + \beta_2 Post\ Covid_t \times Treat^{Exp\ Office}_{jt} + \varepsilon_{ijt}, \end{aligned} \tag{3}$$

where I^{sold}_{ijt} is a dummy which equals 1 if bond j was sold by insurer i in year t, $Post\ Covid_t$ equals 1 after 2019, and $Treat^{Exp\ Office}_{jt}$ equals 1 if bond j is exposed to office-linked mort-

gages whose main lease expires within six years (excluding year t). We do not include delinquent loans when creating the lease expiration treatment dummies so as to avoid capturing the effect of concurrent losses. α_{it}, α_{ij}, $\alpha_{j(coupon)t}$, $\alpha_{ij(NAIC)t}$, and $\alpha_{ij(downgrade_{t-1})t}$ are, respectively, insurer by year, insurer-security, coupon type by year, insurer-specific NAIC designation by year, and lagged insurer-specific downgrade by year fixed effects.[7]

This battery of time-varying fixed effects safeguards that our estimated effect does not stem from portfolio adjustments resulting from shocks to insurers (Ge and Weisbach, 2021), sales driven by downgraded assets, with potentially higher capital requirements, or due to regulatory reasons (Ellul, Jotikasthira and Lundblad, 2011; Becker, Opp and Saidi, 2022).

The results are in Table 3 and indicate that insurers do take into account how bonds are affected by cash flow shocks in the short to medium term when selling CMBS. In column 1, we estimate (3) and find that insurance companies are 2.4 percentage points more likely to sell bonds which have office mortgages that expire in the next six years after the COVID-19 pandemic than before. The mean of the dependent variable I_{ijt}^{sold} equals 8% for private-label CMBS, indicating a meaningful economic effect from exposure to cash flow shocks expected to materialize in the short to medium term.

This effect is robust to other characteristics of the mortgage pool of a given CMBS that could affect the likelihood of selling the bond. In particular, we account for the fact that $Treat_{jt}^{Exp\ Office}$ reflects de facto an interaction between any office exposure and lease expiration within six years, which we control for in turn.

In column 2, we include a dummy variable for the existence of any offices in the underlying pool of mortgages of bond j in year t, I_{jt}^{Office}. In spite of incorporating this control variable along with its interaction with the post-COVID indicator, our estimate remains virtually unaltered. This suggests that the post-COVID increase in sales is concentrated among office-exposed bonds with lease expirations in the short to medium term.

In the same way, we account for lease expirations more generally. In column 3, we include a dummy variable that equals 1 for CMBS with *any* mortgage with a lease expiring within six years starting in year t, I_{jt}^{Exp}. In line with the idea that insurers are sophisticated enough to understand how collateral type and lease characteristics jointly determine expected mort-

[7] We defined downgrading based on NAIC designations. Hence, a bond j held by insurer i is marked as downgraded in year t if it had a NAIC designation in year $t-1$ greater than its NAIC designation in year $t-2$.

gage default, we find that underlying lease expiration affects only the post-COVID sales probability for bonds associated with office collateral. Furthermore, once the additional collateral controls are included, we find no evidence of differential sales associated with lease expiration or office exposure before COVID-19. This lends support to our conjecture that insurance companies learn about the increase in riskiness of the underlying collateral of CMBS posed by work-from-home demand shocks.

Our evidence in Section 4 reveals contrasting patterns between office and retail loan delinquency rates. Loans linked to retail experience a spike in delinquency right at the onset of the pandemic, which would also pose a risk to holders of CMBS exposed to retail properties. Importantly, this risk is less sensitive to lease expiration, suggesting that the latter is less relevant for retail-exposed CMBS in comparison to office-exposed CMBS. Column 4 of Table 3 tests this idea by further breaking down the collateral and the lease-expiration dummies into mutually exclusive property types. In particular, the dummies $I_{jt}^{Exp\ Retail}$ and $I_{jt}^{Exp\ Other}$ equal 1 if bond j has underlying retail or other mortgages, respectively, with leases expiring within six years, where *Other* is a residual category for any loans with lease expiration information not linked to $Office$ or $Retail$ units. Moreover, I_{jt}^{Retail} is a dummy that equals 1 if bond j has any exposure to retail units in year t.

The coefficient of interest, β_2, drops somewhat after adding these refined controls but remains statistically significant, confirming that insurers are sensitive to cash flow risks for offices after COVID-19. Overall, our evidence supports the idea that insurers do not only monitor cash flow risks but are also sufficiently sophisticated to disentangle how these risks affect different types of CMBS collateral.

To assess whether exposure to office lease-rollover risk predicts potential losses to bondholders, we investigate whether affected bonds are more likely to experience interest shortfall. To avoid capturing interest delays caused by COVID-19 lockdowns, we define an alternative post-treatment variable, $Post\ Covid_t^{2021,2022}$, which equals 1 in 2021 and 2022. Using this post-treatment variable, we re-estimate Equation (3), but using $I_{jt}^{Interest\ Shortfall}$, which equals 1 if bond j registered positive interest shortfall in year t, as the dependent variable. Appendix-Table A.4 shows that across most specifications exposure to office lease expiration has predictive power for increases in interest shortfall between 2021–2022. This indicates that the exposure measure predicts subsequent deterioration in bond-level perfor-

mance, consistent with insurers selling ahead of such deterioration.

A more general concern is that the treatment variable, $Treat_{jt}^{Exp\ Office}$, can be correlated with other characteristics that could affect bond liquidity. The summary statistics for treatment and control bonds are in Appendix-Table A.5; they reveal differences between treated and control bonds, most notably in conduit status, as well as differences in subordination, seniority, and DSCR. To ensure our results do not reflect these differences, we show that allowing each major baseline characteristic to predict differential post-COVID sales behavior does not affect the results in Table 3. We do so by re-estimating Equation (3) and including each variable in Appendix-Table A.5 interacted with the $Post\ Covid_t$ dummy. We also estimate a specification including all covariate interactions jointly. Appendix-Table A.6 shows that our main results are virtually identical across all specifications, alleviating concerns that our results reflect other differences between treatment and control bonds that are not related to expected cash flows from office leases.

To bolster our identification assumption that insurers would not have reacted differently to shocks affecting the cash flow risks of CMBS exposed to offices with leases expiring within six years in the absence of the pandemic, we also estimate a dynamic difference-in-differences regression:

$$\begin{aligned} I_{ijt}^{sold} = {} & \alpha_{it} + \alpha_{ij} + \alpha_{j(coupon)t} + \alpha_{ij(NAIC)t} + \alpha_{ij(downgrade_{t-1})t} \\ & + \beta Treat_{jt}^{Exp\ Office} + \sum_{\iota \neq 2019} \delta_\iota Treat_{j\iota}^{Exp\ Office} + \theta Controls_{jt} + \varepsilon_{ijt}, \end{aligned} \tag{4}$$

where each $Treat_{j\iota}^{Exp\ Office}$ is a dummy variable which equals 1 in year $\iota \neq 2019$ if bond j is exposed to office-linked mortgages whose main lease expires within six years (excluding year ι), and $Controls_{jt}$ includes the same set of bond-level controls as in column 3 of Table 3. Figure A.2 plots the estimated coefficients δ_ι. Reassuringly, we find no visual evidence for violation of the parallel trends assumption. Most of the effect we capture takes place in 2020 and in 2022, with a spike in sales of CMBS with more exposure to cash flow risks posed by lease expiration.

To what extent do these sales represent actual portfolio rebalancing, rather than minor adjustments in insurers' asset portfolios? To answer this question, we follow the approach

from Ge and Weisbach (2021), and re-estimate Equation (3) using as dependent variable the par value of the sales of bond j in year t divided by the par value of the same bond in the previous year, $Share\ Par\ Value\ Sold_{ijt}$. The results are in Appendix-Table A.8. Across all specifications, the coefficient of interest β_2 is positive and significant, indicating that after COVID-19 a larger share of bonds exposed to cash flow shocks was sold by insurers, as measured by the share of par value sold. Our findings are robust to this alternative specification and, thus, capture meaningful adjustments in the asset portfolio of insurance companies.

5.3. Intensive-margin Monitoring of Cash Flow Risks

Our baseline treatment provides a transparent and relatively assumption-light indicator of whether a bond is exposed to medium-term office lease-expiration risk. Nonetheless, the measure does not differentiate between the number of office loans with expiring leases in each year, the mortgage balance of the pool exposed to these loans, and the relative size of each tenant with leases expiring. We now use richer tenant-level information to test whether trading responses also increase with the *magnitude* of that exposure.

To capture both the magnitude and the timing of lease-rollover exposure, we construct the present discounted value (PDV) of future lease expirations for each bond's corresponding deal. For each bond j and year t, we first calculate the amount of outstanding collateral exposed to lease expiration in each future year τ. For every loan in the corresponding deal, we use lease information for the five largest tenants and weight the loan's outstanding balance by the share occupied by each tenant whose lease expires in year τ. We then aggregate these tenant-weighted loan balances across all loans in the deal linked to each bond. Formally, future lease-expiration exposure is given by:

$$A_{jt\tau} = \sum_{l \in \mathcal{D}(j)} \sum_{k=1}^{5} B_{ljt} s_{lkt} \mathbb{1}\{E_{lkt} = \tau\}, \tag{5}$$

where B_{ljt} denotes the outstanding balance of loan l in the deal linked to bond j in year t, $s_{lkt} \in [0,1]$ is the share occupied by tenant k, and E_{lkt} is that tenant's lease-expiration year. Thus, $A_{jt\tau}$ measures the amount of bond j's outstanding collateral that is exposed to lease expiration in year τ, accounting for the relative size of each tenant.

We then discount these future exposures using the U.S. Treasury zero-coupon yield curve and normalize their present value by the total outstanding collateral balance of the deal:

$$PDV_{jt} = \frac{\sum_{\tau=t+1}^{t+T} m_{t,\tau} A_{jt\tau}}{B_{jt}}, \tag{6}$$

where $m_{t,\tau}$ is the discount factor associated with an exposure occurring in year τ, T denotes the maximum horizon over which future lease expirations are measured, and $B_{jt} \equiv \sum_{l \in \mathcal{D}(j)} B_{ljt}$ is the total amount outstanding of the deal linked to bond j in year t. Assuming continuous compounding, we calculate the discount factor as:

$$m_{t,\tau} = \exp\left[-y_{t,\tau-t}(\tau - t)\right], \tag{7}$$

where $y_{t,\tau-t}$ denotes the U.S. Treasury zero-coupon yield in year t at maturity $\tau - t$.[8] The resulting PDV_{jt} can be interpreted as the discounted share of a bond's current collateral balance exposed to future lease rollover. We construct the measure separately for office, retail, and also for all properties so as to match the specifications in Table 3. We do so by restricting the set of loans entering Equation (5) to the corresponding property type. Furthermore, using the full term structure of lease-expiration exposures allows us to understand whether bonds with larger exposure to lease-rollover risk are likely to be sold by insurers after COVID-19.

To assess whether trading responses vary with the magnitude of lease-rollover exposure, we first test if bonds in the upper tercile of the distribution of office PDV_{jt} have a higher likelihood of being sold. This is analogous to the baseline difference-in-differences specification using $Treat_{jt}^{ExpOffice}$ but, instead, we use a threshold dummy which equals 1 if the value of PDV_{jt} is in the top tercile of the distribution of the same variable in the sample. This specification allows us to directly assess whether higher PDV of exposure to lease expiration affects the probability of bond sales. The results in Table 4 indicate that higher exposure to lease-rollover risk, as captured by PDV_{jt}, indeed affects the likelihood of bond sales after COVID-19.

We next examine the intensive margin of lease-rollover exposure. Rather than imposing a

[8] The original series is obtained from the Federal Reserve, based on Gurkaynak, Sack and Wright (2007).

linear relationship between the PDV measure and insurers' trading response, we divide the distribution of PDV exposure into terciles. This allows the effect of additional lease-rollover exposure to vary flexibly across the distribution. This flexibility is particularly relevant in our setting because the mapping from collateral exposure to bond-level credit risk is unlikely to be linear: CMBS tranches differ in credit protection, small exposures may have little effect on expected bond losses, and monitoring or portfolio-adjustment decisions may themselves involve economically relevant thresholds. To this end, we estimate the following specification:

$$\begin{aligned} I_{ijt}^{sold} =& \alpha_{it} + \alpha_{ij} + \alpha_{j(coupon)t} + \alpha_{ij(NAIC)t} + \alpha_{ij(downgrade_{t-1})t} + \\ & \beta_1 \times Q2_{jt}^{Exp\ Office} + \beta_2 \times Q3_{jt}^{Exp\ Office} \\ & + \gamma_1 Post\ Covid_t \times Q2_{jt}^{Exp\ Office} + \gamma_2 Post\ Covid_t \times Q3_{jt}^{Exp\ Office} + \varepsilon_{ijt}, \end{aligned} \tag{8}$$

where $Q2_{jt}^{Exp\ Office}$ and $Q3_{jt}^{Exp\ Office}$ correspond to the second and third terciles of the distribution of the PDV office exposure variable. We employ a similar approach as in Table 3, and control for the terciles of the discounted value of retail and other properties' exposure to lease expiration, and the terciles of the bond exposure to offices and retail.

The results in Appendix-Table A.7 show that the likelihood of sales *increases monotonically* with the magnitude of the exposure to lease-rollover risk captured by the PDV measure across all specifications. The top tercile of the exposure measure, $Q3_{jt}^{Exp\ Office}$, consistently predicts a higher likelihood of bond sales after the pandemic at conventional levels of statistical significance.

5.4. Heterogeneity Across Bond and Insurer Characteristics

The effect of expected cash flow shocks should also vary with a bond's position in the CMBS capital structure. Because subordinate tranches absorb losses before senior bonds, their values are more exposed to changes in expected collateral cash flows. As a result, information about future cash-flow deterioration is likely to have larger valuation consequences for junior securities. This is consistent with theories of security design in which claims differ in their sensitivity to information about underlying asset fundamentals. Senior debt-like securities are relatively information-insensitive, whereas more junior claims concentrate expo-

sure to asset-level risk and therefore provide stronger incentives for information acquisition and trading (Gorton and Pennacchi, 1990; DeMarzo and Duffie, 1999). If insurers actively monitor expected cash flow shocks, we therefore expect the positive relation between shock exposure and sales to be strongest among CMBS that are less senior and less protected by subordination.

Roughly 30% of private-label CMBS holdings by insurers in our sample are non-senior tranches (e.g., mezzanine or junior), and the average subordination percentage is also approximately 30%. This cross-sectional variation in seniority and subordination across CMBS held by insurers allows us to evaluate the hypothesis that, among exposed tranches, those that are more informationally sensitive are more likely to be sold. To do so, we employ a triple-difference specification, where we interact our treatment variable with variables that capture differences in seniority and subordination:

$$\begin{aligned} I_{ijt}^{sold} =& \alpha_{it} + \alpha_{ij} + \alpha_{j(coupon)t} + \alpha_{ij(NAIC)t} + \alpha_{ij(downgrade_{t-1})t} \\ &+ \beta_1 \times Treat_{jt}^{Exp\ Office} + \beta_2 Post\ Covid_t \times Treat_{jt}^{Exp\ Office} \\ &+ \beta_3 \times Treat_{jt}^{Exp\ Office} \times X_{jt}^{Senior} + \beta_4 \times Post\ Covid_t \times X_{jt}^{Senior} \\ &+ \beta_5 \times Post\ Covid_t \times Treat_{jt}^{Exp\ Office} \times X_{jt}^{Senior} + \varepsilon_{ijt}, \end{aligned} \tag{9}$$

where X_{jt}^{Senior} is either one of two variables: $I_j^{Non-senior}$, an indicator variable for non-senior tranches, and $Subordination(\%)_{jt}$, the subordination percentage of bond j in year t. The results are in Table 5, using the same range of controls as before.[9]

Both sets of results support the hypothesis that monitoring of expected cash flow losses due to office lease-expiration risk is more intensive for more information-sensitive bonds. Columns 1 through 4 show that non-senior bonds exposed to increased office lease-rollover risk are between 3 and 5 percentage points more likely to be sold post-COVID than senior bonds that are also exposed to office lease-rollover risk. Similarly, the results in columns 5 through 8 indicate that a higher subordination percentage is associated with lower sensitivity to increases in expected cash flow risk.

Another possibility is that insurers' investment expertise also affects their monitoring of

[9]For simplicity, we only display the point estimates for β_1, β_2, and β_5.

CMBS and leads to changes in how they react to the increase in expected cash flow shocks. We test this hypothesis by estimating a variant of Equation (9), in which we set X_i^{Senior} to be equal to one of the following insurance characteristics: the share of each insurer's fixed income portfolio invested in private-label CMBS and in risky assets, the % share of each insurer's private-label CMBS that is invested in non-senior tranches, and the average subordination % of the private-label CMBS tranches held by each insurer.

Appendix-Table A.9 shows that insurers with a higher share of their fixed income portfolio invested in private-label CMBS, and those whose private-label CMBS portfolio is invested in a larger share of non-senior and low subordination bonds, exhibit stronger trading responses to exposed bonds, consistent with more intensive monitoring. This is in line with the idea that insurers with more expertise in private-label CMBS and who invest in more informationally sensitive tranches engage in more intensive monitoring of expected cash flow shocks caused by an increase in office lease-rollover risk.

5.5. CMBS Trading and Realized Gains and Losses

The results so far suggest that insurers actively monitor underlying lease expiration information embedded in complex CMBS structures to identify and sell assets exposed to imminent cash flow shocks. However, the sale of CMBS exposed to such cash flow risks could result in lower revenue and realized losses for insurers, which could reflect distressed sales.

To assess whether this holds in practice, we examine realized gains and losses reported by insurers on their trading activities. These realized gains and losses correspond to the difference between the sale proceeds of an asset and its book value. This approach allows us to capture realized losses relative to BACV—precisely what would occur if insurers had to make forced sales. Using this information, we estimate the following specification:

$$\begin{aligned} Realized\ gain\ over\ BACV_{ijt} = \alpha_{it} + \alpha_j + \alpha_{j(coupon)t} + \alpha_{ij(NAIC)t} + \alpha_{ij(downgrade_{t-1})t} \\ + \beta_1 Treat_{jt}^{Exp\ Office} + \beta_2 Post\ Covid_t \times Treat_{jt}^{Exp\ Office} + \varepsilon_{ijt}, \end{aligned} \tag{10}$$

where the dependent variable, $Realized\ gain\ over\ BACV_{ijt}$, measures realized gains or losses on the sale of bond j by insurer i in year t, relative to the book value of bond j in the previous

year. The treatment variable $Treat_{jt}^{Exp\ Office}$ and the fixed effects α_{it}, $\alpha_{j(coupon)t}$, $\alpha_{ij(NAIC)t}$, and $\alpha_{ij(downgrade_{t-1})t}$ are defined as before, where α_j denotes bond-level fixed effects. As realized gains and losses are observed for bonds that are sold, we estimate (10) using only bonds sold, compromising our ability to include insurer-bond fixed effects α_{ij}, which would require multiple sales of the same bond by a given insurer.

The coefficient of interest, β_2, captures the differential realized gain or loss for treated bonds (those exposed to office lease-expiration risk) sold in a given year relative to non-exposed bonds sold in the same period. If insurers realized losses when selling exposed bonds, we would expect a *negative* estimated β_2 as the proceeds received by insurers for these assets would be lower than those for comparable non-exposed assets, controlling for other price-relevant characteristics such as ratings downgrades.

Table 6 shows the results. Across all specifications, we find no evidence that treated bonds were sold at a discount after COVID-19. If anything, exposure to office collateral is associated with *better* realized trading performance post–COVID-19, whereas exposure to retail collateral is associated with *negative* outcomes. This pattern underscores that insurers' CMBS trading gains and losses do respond to salient features of collateral performance—consistent with the divergent delinquency dynamics observed for offices and retail properties at the onset of the pandemic (Figure 2). In contrast, we find no systematic relationship between exposure to *lease-expiration risk* in offices and realized gains or losses on insurers' securities sales.

Finally, we also provide evidence that insurers react to expected cash flow risks in their *acquisitions* of office-exposed CMBS after COVID. In Appendix-Table A.10, we show that coupons of newly issued CMBS with exposure to offices command higher coupons when these tranches are acquired by insurance companies. These changes in acquisition behavior by insurers further corroborate that they monitor changes in office loan risk associated with work-from-home arrangements.

6. Bank Holdings of Private-label CMBS

Our evidence suggests that insurers are able to monitor risks in securitized assets that arise from lower office demand after the pandemic, and reduce their exposure to these private-

label CMBS. In this section, we further document patterns of CMBS holdings by examining banks' securities portfolios and the realized gains and losses banks incur on their securities trading. We use Call Reports data and construct bank-level holdings of private-label CMBS.

In this manner, we first document how aggregate holdings of private-label CMBS evolve over time for banks of different sizes. Figure A.3 shows a noticeable increase in holdings of CMBS by small and medium-sized banks (i.e., those with assets under $100 billion) from 2021 onwards. This pattern is more striking relative to 2017 and 2018, when the aggregate amount of private-label CMBS holdings by banks was at similar levels but with a substantially smaller role played by small and medium-sized banks.

Figure 6 focuses on small banks, and documents two important patterns. In Panel A, we see that the number of small banks (total assets under $10 billion) that hold private-label CMBS nearly doubles between March 2020 and December 2023. Moreover, these "new entrants" are holding meaningful shares of private-label CMBS. Panel B shows that small banks' total holdings of private-label CMBS relative to non-Treasury, non-agency securities increases sharply between 2021 and 2023. This indicates that the increase in small banks' investment in private-label CMBS in Figure A.3 is not solely due to the expansion of banks' securities portfolios.

To understand how this unprecedented increase in the number of small banks investing in CMBS is related to the characteristics of these banks, we divide our sample of small banks into three types: banks that held private-label CMBS between 2017 and March 2020, banks that held private-label CMBS *only after* March 2020, and banks that did not hold any private-label CMBS between 2017 and 2023. Appendix-Table A.11 shows mean values of selected characteristics for these three different bank types. Banks that began investing in private-label CMBS after COVID-19 are smaller than those that invested in CMBS before the pandemic, but have similar leverage, exposure to CRE loans, and have a similar share of their total assets and securities invested in private-label CMBS. Banks that did not invest in CMBS (column 3) are smaller, have lower exposure to non-owner-occupied CRE loans, and are less levered than banks that invested in private-label CMBS between 2017 and 2023.

The increase in private-label CMBS holdings by small banks may be the result of COVID support policies, subsequent deposit inflows, changes in risk appetite, or a response to changes in monetary policy. Nevertheless, the growing presence of small banks in private-label CMBS

markets raises the possibility that these institutions subsequently face losses in their securities portfolios. To assess this hypothesis, we examine realized gains and losses on banks' securities holdings, comparing institutions exposed and not exposed to private-label CMBS. We estimate the following specification:

$$\frac{Securities\ Losses_{it}}{Assets_{it-1}} = \alpha_i + \alpha_t + \beta_1 Treat_{it-1} + \beta_2 X_{it} + \sum_{\tau \neq 2020} \delta_\tau Treat_{i\tau-1} + \sum_{\tau \neq 2020} \Gamma_\tau \times X_{i\tau} + \varepsilon_{it}, \quad (11)$$

where the dependent variable, $\frac{Securities\ Losses_{it}}{Assets_{it-1}}$, measures realized gains or losses on held-to-maturity and available-for-sale securities of bank i in year t, scaled by total assets in the previous year and expressed in percentage points. The treatment indicator, $Treat_{it-1}$, equals 1 if bank i held private-label CMBS in year $t-1$. The specification includes bank (α_i) and year (α_t) fixed effects to absorb time-invariant heterogeneity across banks and aggregate shocks common to all institutions. Standard errors are clustered at the bank level.

We further include a vector of time-varying bank-level controls, X_{it}, to account for other factors influencing securities performance but unrelated to CMBS exposure. These controls include lagged shares of (i) treasury securities, (ii) agency MBS, (iii) long-duration assets, and (iv) short-duration assets, all out of total securities held. We also include total securities relative to total assets as well as lagged total assets as controls. We further allow the importance of each characteristic to vary across years by estimating time-varying coefficients.

The coefficients of interest, δ_τ, capture differences in securities portfolio performance between banks that hold private-label CMBS and those that do not in the year prior, relative to the baseline period (late 2020). The results, reported in Figure 7, reveal a pronounced relative decline in realized gains and losses for banks exposed to private-label CMBS in 2022 and 2023. The timing of this decline coincides with the stabilization in the number of small banks holding private-label CMBS (Figure 6). In 2022-2023, treated banks experience approximately 6 basis points lower realized performance in their securities portfolios (relative to total assets) compared with non-exposed banks. Given that the ratio of Tier 1 capital to total assets for small banks averages around 11%, this loss corresponds to roughly a 1% relative decline in Tier 1 capital for banks holding private-label CMBS aggregated over the two years. We also find no evidence of differential pre-trends prior to the pandemic, supporting the validity of our design.

It is important to caveat, however, that we cannot attribute these portfolio losses specifically to CMBS credit deterioration. Other factors, such as valuation declines resulting from higher interest rates, along with other losses in banks' securities portfolios, can affect realized gains and losses in ways that are not captured by the controls included in Equation (11).

7. Conclusion

In this paper, we examine how cash flow shocks from commercial property leases affect mortgage delinquencies, and assess insurers' monitoring of these risks in securitized assets. Using detailed data on commercial mortgages in CMBS deals combined with insurers' portfolio and trading behavior, we document that lease expiration becomes a stronger predictor of CRE mortgage delinquency following COVID-19. For office properties specifically, lease expiration becomes a stronger predictor of default during the pandemic, likely driven by reduced office demand due to work-from-home arrangements.

Our analysis reveals that insurers actively manage these risks by reducing exposure to vulnerable CMBS before delinquencies materialize. Moreover, in line with theories of securities' information sensitivity, the reduction in exposure to vulnerable CMBS is more pronounced for non-senior tranches and tranches with less subordination protection. Importantly, these sales do not occur at a loss, in contrast to sales resulting from ratings downgrades. We also document a concurrent increase in CMBS holdings by small banks, raising their overall exposure to CRE risks.

Taken together, our results show that institutional investors can process granular information from collateral pools of complex securitized assets, and respond to changes in expected risks before losses materialize. The findings highlight active collateral monitoring as an important dimension of insurers' risk management beyond responses to credit ratings, regulation, and other security-level signals.

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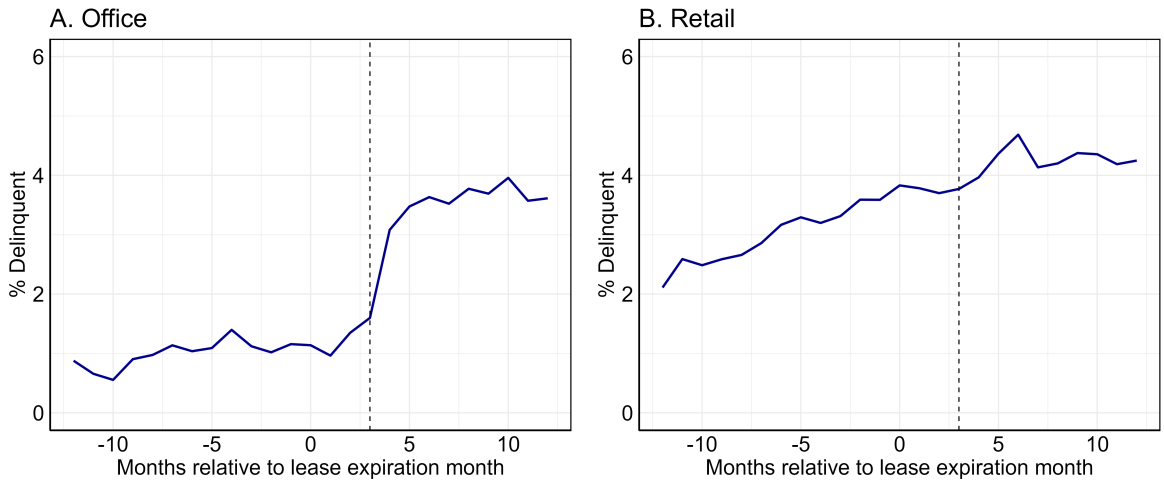
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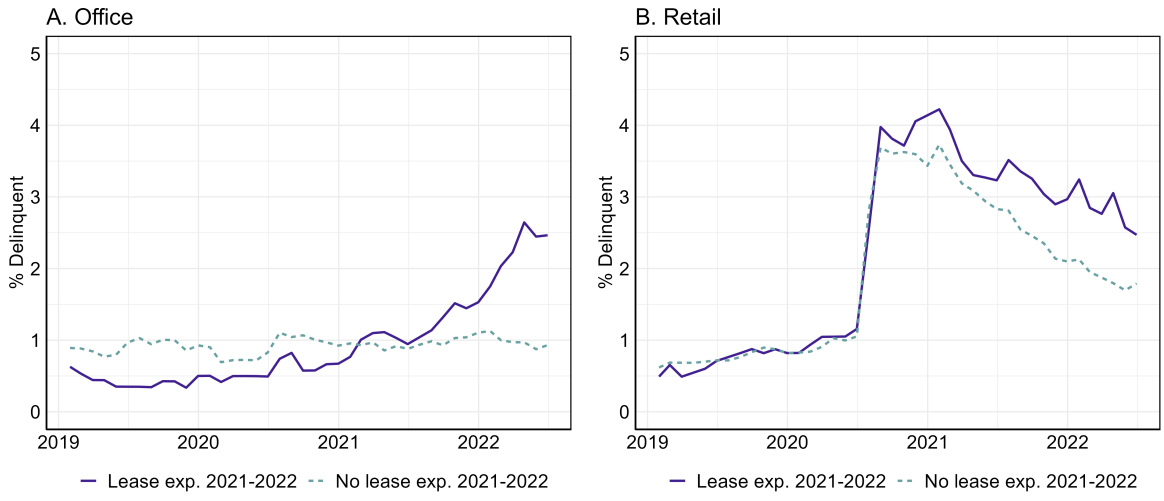
FIGURES

Figure 1: Changes in Delinquency Around Lease Expiration Dates for Office and Retail



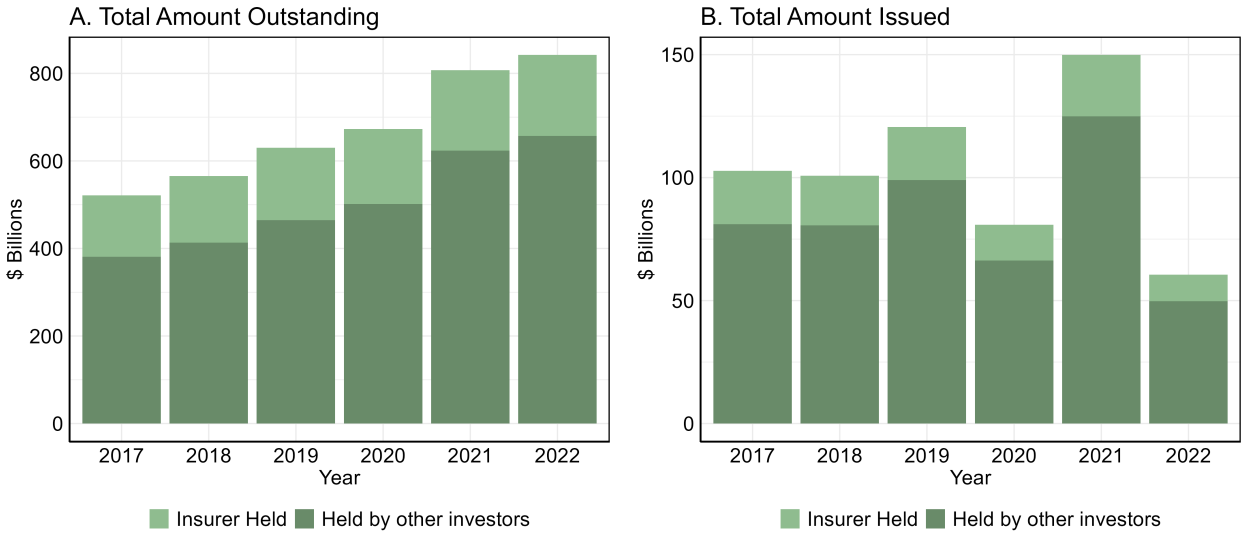
Notes: This figure shows average delinquency in each month relative to lease expiration, for properties with leases expiring between 2017 and June 2022. **Panel A** shows delinquency rates for properties classified as *Office*. **Panel B** shows delinquency rates for properties classified as *Retail*. Delinquency is a dummy variable which equals 1 if a mortgage is at least 90 days past due. The vertical line marks three months after lease expiration. Sources: Trepp loan data and authors' calculations.

Figure 2: Delinquency Rates of Mortgages With and Without Leases Expiring in 2021-2022



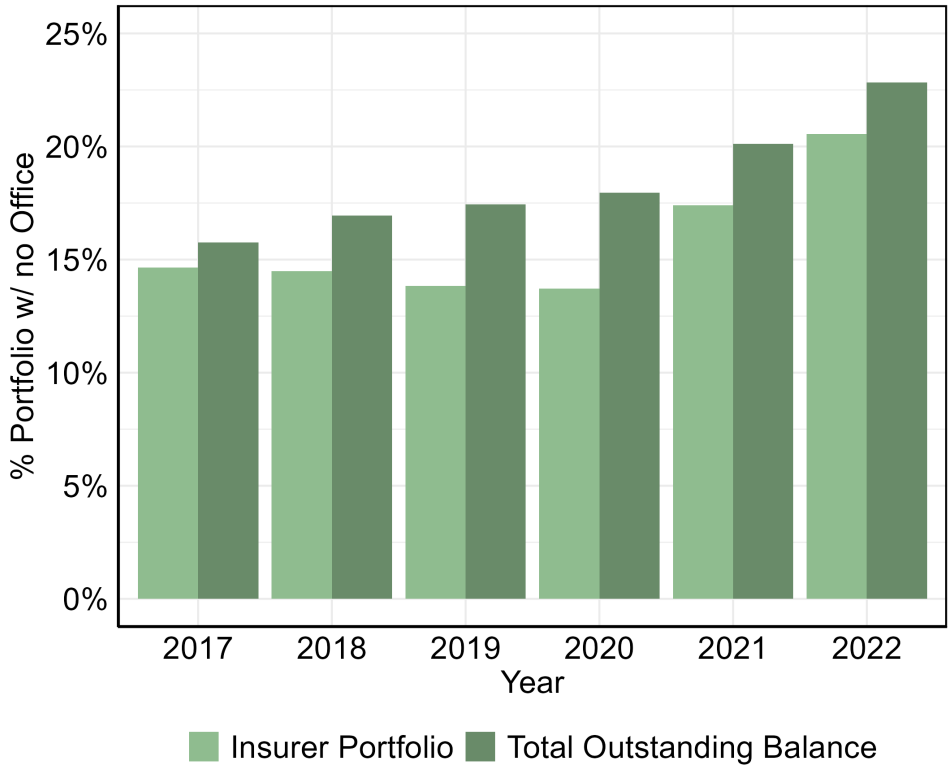
Notes: This figure shows average delinquency rates for mortgages *with* leases expiring in 2021-2022, and mortgages *without* leases expiring in these two years. **Panel A** shows delinquency rates for properties classified as *Office*. **Panel B** shows delinquency rates for properties classified as *Retail*. Delinquency is a dummy variable which equals 1 if a mortgage is at least 90 days past due. Sources: Trepp loan data and authors' calculations.

Figure 3: Insurance Holdings of CMBS



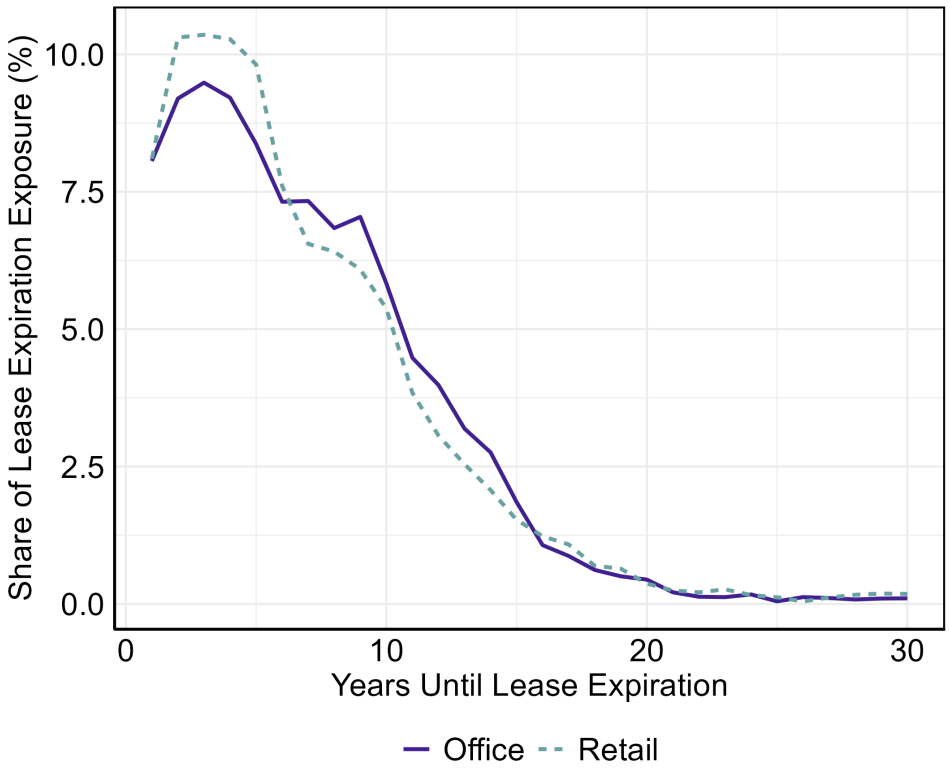
Notes: This figure shows the total amount outstanding (**Panel A**) and amount originated (**Panel B**) of private-label CMBS per year, differentiating between the amount held by insurance companies and that held by other investors. We identify holdings of insurance companies using NAIC Schedule D, Part 1. Insurer-held amounts are calculated as the sum of the BACV of the CMBS held by insurers. Amount held by other investors is the residual value relative to the total original balance outstanding/originated in a given year. Both plots exclude interest-only and agency CMBS. Source: Trepp, NAIC, and authors' calculations.

Figure 4: % CMBS Portfolio Without Office Exposure



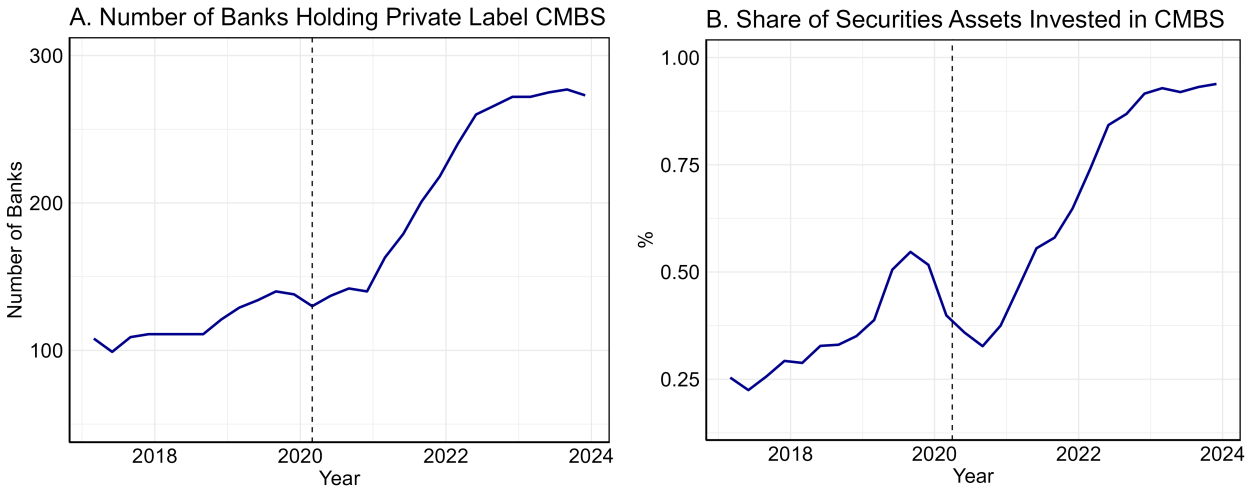
Notes: This figure shows the share of insurance companies' private-label CMBS portfolio, and the share of the total outstanding balance of CMBS, which are not exposed to *any* mortgages linked to properties classified as *Office*. Shares are calculated aggregating BACV for exposed and non-exposed CMBS for insurers' portfolio, and based on outstanding balances for the market-wide numbers. Exposure is defined as any percentage of the pool of mortgages used to finance office CRE. Source: Trepp, NAIC, and authors' calculations.

Figure 5: CMBS Cash Flow Shock Dynamics



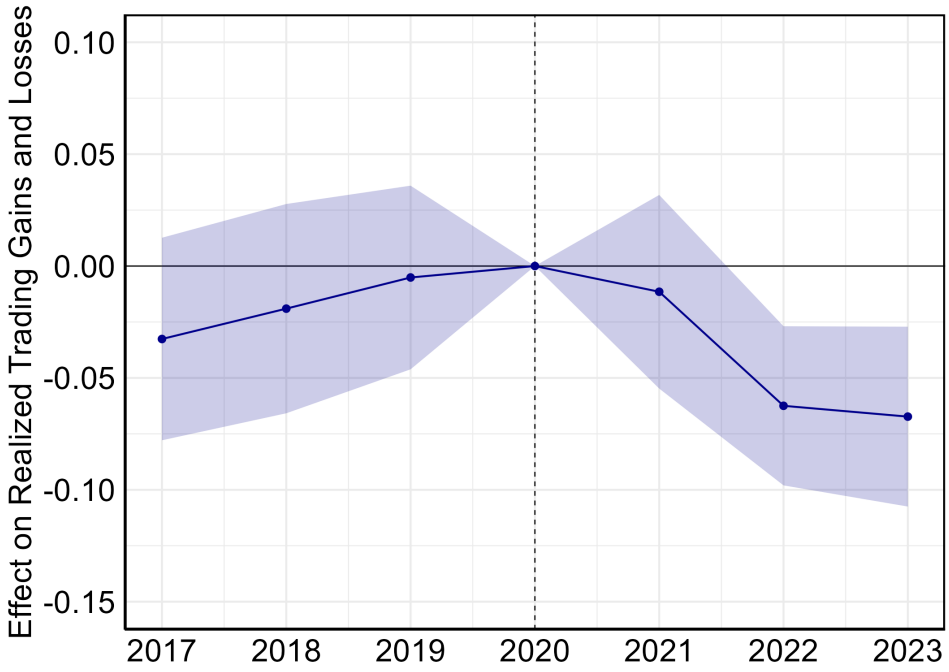
Notes: This figure shows the average distribution of expected cash flow shocks for deals in our sample in each lease-expiration year after the reference year. We first calculate, for each deal in a given year, what share of mortgages have leases expiring in τ years. Then, we average these shares, first across bonds of a given property type in a given year, then across years, separately for offices and retail. Source: Trepp, NAIC, and authors' calculations.

Figure 6: Small Banks' Exposure to Private-Label CMBS



Notes: This figure shows the number of small U.S. banks which hold private-label CMBS (Panel A) and the ratio of small banks' private-label CMBS holdings to a broader portfolio of non-Treasury, non-agency securities, expressed as a percentage (Panel B). Small banks are defined as institutions with total assets under $10 billion. Source: Call Reports and authors' calculations.

Figure 7: Bank Holdings of CMBS and Securities Losses



Notes: The plot shows the dynamic effect of banks' exposure to private-label CMBS on realized gains and losses on securities trading, as in (11). Shaded area corresponds to the 95 percent confidence interval around point estimates. Standard errors are clustered at the bank level. Source: Call Reports and authors' calculations.

TABLES

Table 1: Summary Statistics

Panel A. All Properties	Mean	Median	Min	Max	N
Outstanding Balance	12,126,498.41	4,665,665.69	535.94	9,016,115,069.00	7,081,912
Floating Interest Rate	0.12	0.00	0.00	1.00	7,081,912
Delinquency (90 days)	0.01	0.00	0.00	1.00	7,081,912
Recourse Loan	0.01	0.00	0.00	1.00	7,081,912
Loan Term	228.39	120	1	515	7,010,744
Lease Expiration Year	2026	2024	2016	2099	747,189
Largest Tenant % Sqr Ft	45.11	33.44	0.00	100.00	748,523
Panel B. Office	Mean	Median	Min	Max	N
Outstanding Balance	35,592,214.49	17,545,061.17	6,760.18	3,000,000,000.00	276,561
Floating Interest Rate	0.08	0.00	0.00	1.00	276,561
Delinquency (90 days)	0.01	0.00	0.00	1.00	276,561
Recourse Loan	0.02	0.00	0.00	1.00	276,561
Loan Term	112.87	120	1	363	275,307
Lease Expiration Year	2025	2024	2016	2099	209,965
Largest Tenant % Sqr Ft	42.20	29.71	0.00	100.00	211,306
Panel C. Retail	Mean	Median	Min	Max	N
Outstanding Balance	17,123,979.19	7,331,549.50	797.55	2,400,000,000.00	516,328
Floating Interest Rate	0.02	0.00	0.00	1.00	516,328
Delinquency (90 days)	0.02	0.00	0.00	1.00	516,328
Recourse Loan	0.01	0.00	0.00	1.00	516,328
Loan Term	123.63	120	1	360	506,734
Lease Expiration Year	2027	2024	2016	2099	415,663
Largest Tenant % Sqr Ft	45.68	34.52	0.00	100.00	417,838

Notes: This table shows summary statistics from our sample of commercial real estate mortgages, across the loan-month sample. The sample period is from Jan/2017 to Jun/2022. **Panel A** includes summary statistics for all property types in the sample. **Panel B** includes summary statistics for properties classified as *Office*. **Panel C** includes summary statistics for properties classified as *Retail*. Source: Trepp and authors' calculations.

Table 2: The Effect of Lease Expiration on Mortgage Default Before and After COVID-19

	I_{jrt}^{D90}		
	(1)	(2)	(3)
$Post\ Expiration_{jt}$	0.0131***	0.0132***	0.0140***
	(0.0029)	(0.0029)	(0.0033)
$Post\ Covid_t \times Post\ Expiration_{jt}$	-0.0029	-0.0029	-0.0013
	(0.0033)	(0.0033)	(0.0038)
$Post\ Covid_t \times Ind\ Office_j$	-0.0162***	-0.0160***	-0.0216***
	(0.0019)	(0.0019)	(0.0030)
$Post\ Expiration_{jt} \times Ind\ Office_j$	0.0013	0.0014	0.0004
	(0.0062)	(0.0062)	(0.0065)
$Post\ Covid_t \times Post\ Expiration_{jt} \times Ind\ Office_j$	0.0122*	0.0121*	0.0132*
	(0.0064)	(0.0064)	(0.0070)
Observations	842,875	842,875	761,489
R^2	0.42919	0.42924	0.52736
Month-year fixed effects	✓		
Loan ID fixed effects	✓	✓	✓
Month-year × Floating fixed effects		✓	✓
Month-year × City fixed effects			✓

Notes: This table shows the effects of lease expiration on delinquency rates for mortgages linked to different property types, before and after COVID-19, as in (1). The level of observation is loan j in city r in year-month t, which refers to the distribution month of each securitized mortgage. The sample period is Jan/2017 to Jun/2022. The dependent variable, I_{jrt}^{D90}, is a dummy variable which equals 1 if a loan is at least 90 days past due. $Post\ Covid_t$ equals 1 after March 2020, $Post\ Expiration_{jt}$ equals 1 if loan j had its main lease expiration before or in year-month t, and $Ind\ Office_j$ equals 1 if loan j is linked to an office. Standard errors clustered at the loan level in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels. Sources: Trepp and authors' calculations.

Table 3: CMBS Trading Difference-in-Differences—Exposure to Lease Expiration

	I^{sold}_{ijt}			
	(1)	(2)	(3)	(4)
$Treat^{Exp\ Office}_{jt}$	0.0092**	0.0103**	0.0015	0.0044
	(0.0041)	(0.0042)	(0.0064)	(0.0053)
$Post\ Covid_t \times Treat^{Exp\ Office}_{jt}$	0.0242***	0.0213***	0.0316***	0.0171***
	(0.0044)	(0.0053)	(0.0079)	(0.0065)
I^{Office}_{jt}		0.0042	0.0022	0.0079
		(0.0173)	(0.0174)	(0.0174)
$Post\ Covid_t \times I^{Office}_{jt}$		0.0064	0.0059	0.0042
		(0.0072)	(0.0072)	(0.0072)
I^{Exp}_{jt}			0.0123*	
			(0.0069)	
$Post\ Covid_t \times I^{Exp}_{jt}$			-0.0134	
			(0.0082)	
$I^{Exp\ Retail}_{jt}$				0.0087
				(0.0063)
$I^{Exp\ Other}_{jt}$				0.0070
				(0.0044)
I^{Retail}_{jt}				-0.0143
				(0.0185)
$Post\ Covid_t \times I^{Exp\ Retail}_{jt}$				0.0069
				(0.0086)
$Post\ Covid_t \times I^{Exp\ Other}_{jt}$				0.0031
				(0.0047)
$Post\ Covid_t \times I^{Retail}_{jt}$				0.0015
				(0.0078)
Observations	206,063	206,063	206,063	206,063
R^2	0.52958	0.52958	0.52961	0.52969
Combined Fixed Effects	✓	✓	✓	✓

Notes: This table shows the effect of exposure to underlying lease expiration and offices on the likelihood of sales of private-label CMBS by insurance companies, as in (3). The level of observation is bond j held by insurer i at the end of year t. The sample period is 2017 to 2022. The dependent variable, I^{sold}_{ijt}, is a dummy which equals 1 if bond j was sold by insurer i in year t. $Post\ Covid_t$ equals 1 after 2019. $Treat^{Exp\ Office}_{jt}$ is a dummy which equals 1 if bond j is exposed to office mortgages whose main lease expires within six years (excluding year t). I^{Exp}_{jt}, $I^{Exp\ Retail}_{jt}$, and $I^{Exp\ Other}_{jt}$ are dummy variables which equal 1 if bond j is exposed to mortgages whose main lease expires within six years (excluding year t), for all properties, retail only, and properties that are neither retail nor offices. I^{Office}_{jt} and I^{Retail}_{jt} are dummies which equal 1 for any exposure to offices and retail. The combined fixed effects include $Year \times InsurerID$, $CUSIP \times InsurerID$, $Year \times Coupon\ Type$, $Year \times NAIC\ Designation$ and $Year \times Lagged\ Downgrade$ fixed effects. Coupon Type is the type of coupon payment for bond j (e.g., fixed rate, floating rate, interest only). Lagged downgrade is a dummy which equals 1 if the bond was downgraded (based on NAIC designation) in year $t-1$. Standard errors clustered at the security level in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels. Sources: Trepp, NAIC, and authors' calculations.

Table 4: CMBS Trading—High PDV of Office Lease-Expiration Exposure

	I_{ijt}^{sold}			
	(1)	(2)	(3)	(4)
$I_{jt}^{High\ PDV,Office}$	0.0084**	0.0083**	0.0068*	0.0069*
	(0.0034)	(0.0036)	(0.0038)	(0.0036)
$Post\ Covid_t \times I_{jt}^{High\ PDV,Office}$	0.0071**	0.0083**	0.0101**	0.0090**
	(0.0034)	(0.0040)	(0.0043)	(0.0040)
$I_{jt}^{Large\ Share,Office}$		-0.0098	-0.0099	-0.0102
		(0.0061)	(0.0061)	(0.0063)
$Post\ Covid_t \times I_{jt}^{Large\ Share,Office}$		-0.0022	-0.0022	0.0018
		(0.0041)	(0.0041)	(0.0043)
$I_{jt}^{High\ PDV,All}$			0.0031	
			(0.0030)	
$Post\ Covid_t \times I_{jt}^{High\ PDV,All}$			-0.0037	
			(0.0038)	
$I_{jt}^{High\ PDV,Retail}$				0.0039
				(0.0038)
$I_{jt}^{High\ PDV,Other}$				0.0066*
				(0.0034)
$I_{jt}^{Large\ Share,Retail}$				0.0017
				(0.0071)
$Post\ Covid_t \times I_{jt}^{High\ PDV,Retail}$				0.0028
				(0.0039)
$Post\ Covid_t \times I_{jt}^{High\ PDV,Other}$				0.0087***
				(0.0033)
$Post\ Covid_t \times I_{jt}^{Large\ Share,Retail}$				0.0060
				(0.0043)
Observations	206,063	206,063	206,063	206,063
R^2	0.52935	0.52938	0.52938	0.52954
Combined Fixed Effects	✓	✓	✓	✓

Notes: This table shows the effect of exposure to underlying lease expiration and offices on the likelihood of sales of private-label CMBS by insurance companies. The level of observation is bond j held by insurer i at the end of year t. The sample period is 2017 to 2022. The dependent variable, I_{ijt}^{sold}, is a dummy which equals 1 if bond j was sold by insurer i in year t. $Post\ Covid_t$ equals 1 after 2019. $I_{jt}^{High\ PDV,Office}$ equals 1 if bond j's present discounted value of exposure to lease expiration of offices is in the upper tercile of our private-label CMBS sample. $I_{jt}^{High\ PDV,All}$, $I_{jt}^{High\ PDV,Retail}$ and $I_{jt}^{High\ PDV,Other}$ are dummy variables of upper terciles of PDV of exposure to lease expiration of all properties, retail and non-office, non-retail properties, respectively. $I_{jt}^{Large\ Share,Office}$, $I_{jt}^{Large\ Share,Retail}$ equal 1 if bond j's % share of office/retail collateral is in the upper tercile of our private-label CMBS sample, respectively. The combined fixed effects include $Year \times InsurerID$, $CUSIP \times InsurerID$, $Year \times Coupon\ Type$, $Year \times NAIC\ Designation$ and $Year \times Lagged\ Downgrade$ fixed effects. Coupon Type is the type of coupon payment for bond j (e.g., fixed rate, floating rate, interest only). Lagged downgrade is a dummy which equals 1 if the bond was downgraded (based on NAIC designation) in year $t-1$. Standard errors clustered at the security level in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels. Sources: Trepp, NAIC, and authors' calculations.

Table 5: CMBS Trading Difference-in-Differences—Exposure to Lease Expiration and Bond Seniority

	I_{ijt}^{sold}							
	Seniority				Subordination			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$Treat_{jt}^{Exp\ Office}$	0.0225***	0.0239***	0.0124*	0.0142**	-0.0082	0.0008	-0.0009	-0.0082
	(0.0050)	(0.0051)	(0.0073)	(0.0063)	(0.0091)	(0.0094)	(0.0138)	(0.0119)
$Post\ Covid_t \times Treat_{jt}^{Exp\ Office}$	0.0126**	0.0087	0.0173*	0.0053	0.0803***	0.0644***	0.0744***	0.0576***
	(0.0056)	(0.0068)	(0.0094)	(0.0080)	(0.0105)	(0.0123)	(0.0185)	(0.0160)
$Post\ Covid_t \times Treat_{jt}^{Exp\ Office} \times I_j^{Non-Senior}$	0.0328***	0.0379***	0.0478***	0.0377***				
	(0.0088)	(0.0107)	(0.0172)	(0.0139)				
$Post\ Covid_t \times I_{jt}^{Office}$		0.0090	0.0080	0.0044		0.0380**	0.0362**	0.0403***
		(0.0104)	(0.0104)	(0.0104)		(0.0149)	(0.0150)	(0.0152)
$Post\ Covid_t \times I_{jt}^{Exp}$			-0.0105				-0.0126	
			(0.0103)				(0.0170)	
$Post\ Covid_t \times I_{jt}^{Exp\ Retail}$				0.0070				-0.0146
				(0.0106)				(0.0184)
$Post\ Covid_t \times I_{jt}^{Exp\ Other}$				0.0087*				0.0010
				(0.0050)				(0.0146)
$Post\ Covid_t \times I_{jt}^{Retail}$				-0.0053				0.0362**
				(0.0106)				(0.0170)
$Post\ Covid_t \times Treat_{jt}^{Exp\ Office}\ Sub(\%)_{jt} \times$					-0.0018***	-0.0014***	-0.0015***	-0.0013***
					(0.0003)	(0.0004)	(0.0005)	(0.0005)
Observations	206,063	206,063	206,063	206,063	191,136	191,136	191,136	191,136
R^2	0.52975	0.52976	0.52980	0.52994	0.53229	0.53235	0.53236	0.53253
Combined Fixed Effects	✓	✓	✓	✓	✓	✓	✓	✓

Notes: This table shows the effect of exposure to underlying lease expiration and offices on the likelihood of sales of private-label CMBS by insurance companies, as in (9). The level of observation is bond j held by insurer i at the end of year t. The sample period is 2017 to 2022. The dependent variable, I_{ijt}^{sold}, is a dummy which equals 1 if bond j was sold by insurer i in year t. $Post\ Covid_t$ equals 1 after 2019. $Treat_{jt}^{Exp\ Office}$ is a dummy which equals 1 if bond j is exposed to office mortgages whose main lease expires within six years (excluding year t). I_{jt}^{Exp}, $I_{jt}^{Exp\ Retail}$, and $I_{jt}^{Exp\ Other}$ are dummy variables which equal 1 if bond j is exposed to mortgages whose main lease expires within six years (excluding year t), for all properties, retail only, and properties that are neither retail nor offices. I_{jt}^{Office} and I_{jt}^{Retail} are dummies which equal 1 for any exposure to offices and retail. The combined fixed effects include $Year \times InsurerID$, $CUSIP \times InsurerID$, $Year \times Coupon\ Type$, $Year \times NAIC\ Designation$ and $Year \times Lagged\ Downgrade$ fixed effects. Coupon Type is the type of coupon payment for bond j (e.g., fixed rate, floating rate, interest only). Lagged downgrade is a dummy which equals 1 if the bond was downgraded (based on NAIC designation) in year $t-1$. Standard errors clustered at the security level in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels. Sources: Trepp, NAIC, and authors' calculations.

Table 6: Insurers' CMBS Trading and Realized Gains and Losses

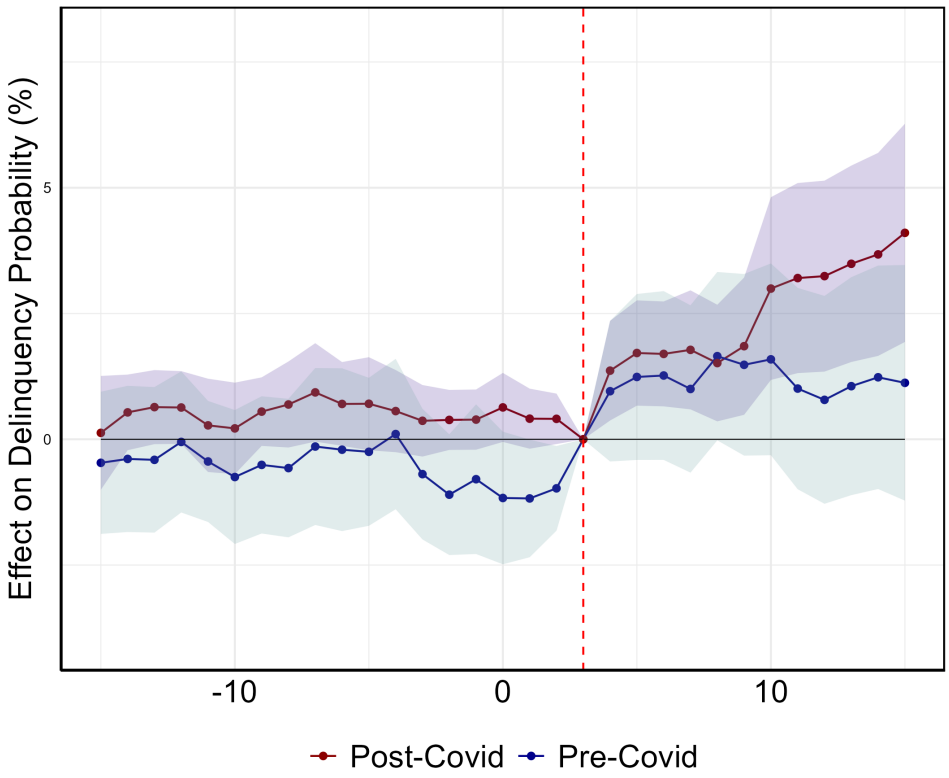
	Realized gain over $BACV_{ijt}$			
	(1)	(2)	(3)	(4)
$Treat_{jt}^{Exp\ Office}$	-0.0015	0.0024	-0.0027	-0.0013
	(0.0041)	(0.0038)	(0.0047)	(0.0041)
$Post\ Covid_t \times Treat_{jt}^{Exp\ Office}$	0.0108**	0.0038	0.0098	0.0087*
	(0.0043)	(0.0044)	(0.0063)	(0.0049)
I_{jt}^{Office}		-0.0277**	-0.0279**	-0.0232*
		(0.0138)	(0.0137)	(0.0134)
$Post\ Covid_t \times I_{jt}^{Office}$		0.0152**	0.0148**	0.0073
		(0.0064)	(0.0064)	(0.0059)
I_{jt}^{Exp}			0.0093	
			(0.0060)	
$Post\ Covid_t \times I_{jt}^{Exp}$			-0.0087	
			(0.0069)	
$I_{jt}^{Exp\ Retail}$				0.0094
				(0.0062)
$I_{jt}^{Exp\ Other}$				-0.0001
				(0.0032)
I_{jt}^{Retail}				0.0190*
				(0.0110)
$Post\ Covid_t \times I_{jt}^{Exp\ Retail}$				0.0087
				(0.0084)
$Post\ Covid_t \times I_{jt}^{Exp\ Other}$				0.0037
				(0.0030)
$Post\ Covid_t \times I_{jt}^{Retail}$				-0.0233***
				(0.0080)
Observations	13,977	13,977	13,977	13,977
R^2	0.78941	0.78975	0.78989	0.79058
Year-Insurer ID fixed effects	✓	✓	✓	✓
CUSIP fixed effects	✓	✓	✓	✓
Year-Coupon Type fixed effects	✓	✓	✓	✓
Year-NAIC Designation fixed effects	✓	✓	✓	✓
Year-Lagged Downgrade fixed effects	✓	✓	✓	✓

Notes: This table shows the effect of exposure to underlying lease expiration and offices on reported gains and losses conditional on bond sales by insurers, as in (10). The level of observation is bond j held by insurer i at the end of year t. The sample period is 2017 to 2022, and includes only bonds which were sold in each given year. The dependent variable, *Realized gain over* $BACV_{ijt}$, is the ratio of realized gains and losses in a given year, divided by the book value of the asset in the previous year. We exclude outliers of this dependent variable below the 2% and above the 98% percentiles. $Post\ Covid_t$ equals 1 after 2019. $Treat_{jt}^{Exp\ Office}$ is a dummy which equals 1 if bond j is exposed to office mortgages whose main lease expires within six years (excluding year t). I_{jt}^{Exp}, $I_{jt}^{Exp\ Retail}$, and $I_{jt}^{Exp\ Other}$ are dummy variables which equal 1 if bond j is exposed to mortgages whose main lease expires within six years (excluding year t), for all properties, retail only, and properties that are neither retail nor offices. I_{jt}^{Office} and I_{jt}^{Retail} are dummies which equal 1 for any exposure to offices and retail. Coupon Type is the type of coupon payment for bond j (e.g., fixed rate, floating rate, interest only). Lagged downgrade is a dummy which equals 1 if the bond was downgraded (based on NAIC designation) in year $t-1$. Standard errors clustered at the security level in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels. Sources: Trepp, NAIC, and authors' calculations.

Internet Appendix

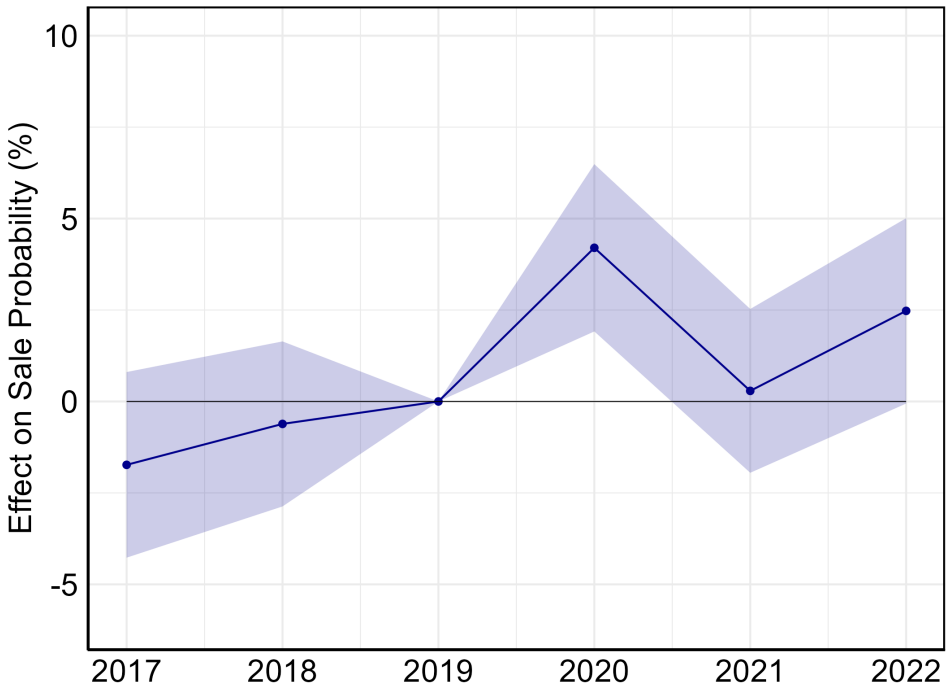
A. Additional Figures and Tables

Figure A.1: Delinquency Rates Around Lease Expiration Dates—Office WFH Sensitivity



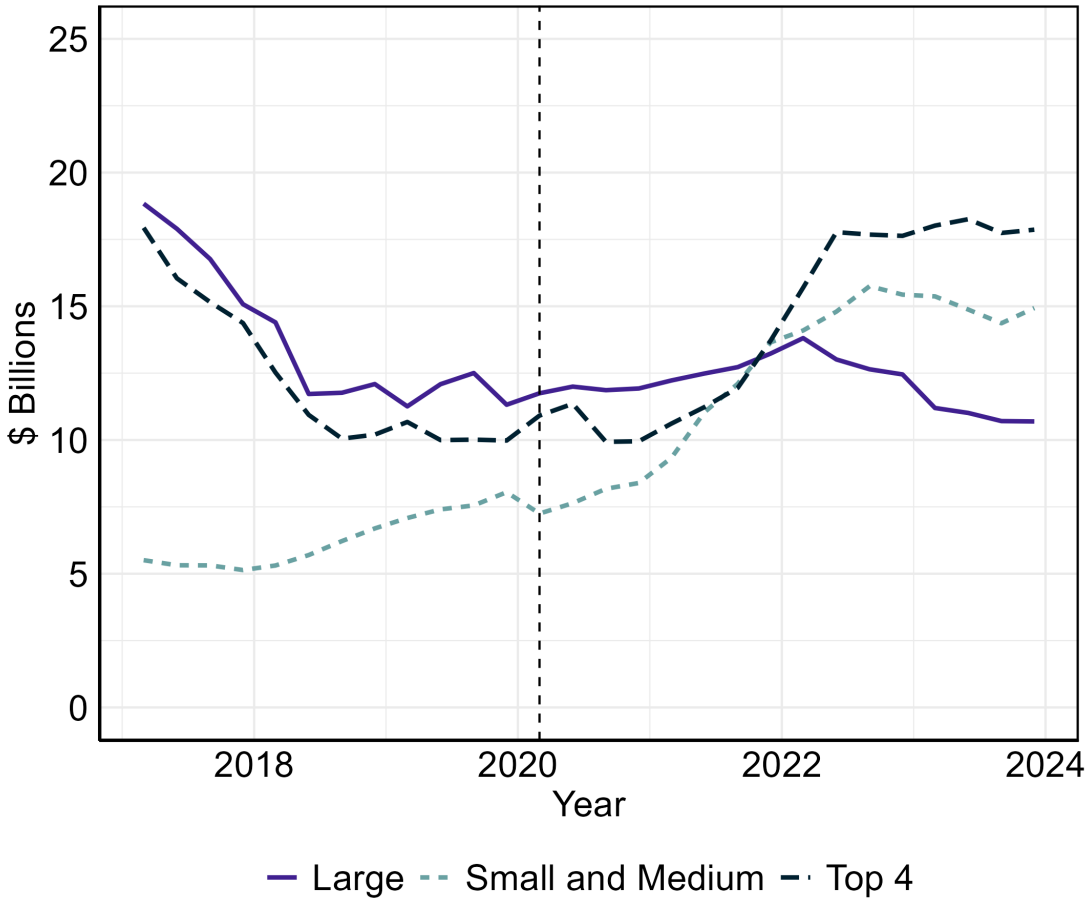
Notes: This figure shows the effects of lease expiration on delinquency rates of properties classified as *Office*. The level of observation is loan j in city r in year-month t, which refers to the distribution month of each securitized mortgage. We estimate a dynamic difference-in-differences specification: $I_{jrt}^{D90} = \alpha_j + \alpha_{rt} + \alpha_{j(floating)t} + \sum_{\iota \in [-15,15] \setminus \{3\}} \delta_\iota D_{jt}^\iota + \varepsilon_{jrt}$. The sample period is Jan/2017 to Jun/2022. The dependent variable I_{jrt}^{D90} is a dummy variable which equals 1 if a loan is at least 90 days past due. The δ_ι estimates show delinquency rates relative to three months after the lease expiration month. α_j and α_{rt} are loan and city by year-month fixed effects. $\alpha_{j(floating)t}$ are interest rate type by year-month fixed effects. The vertical line marks three months after lease expiration. "Pre-Covid" includes all months before March 2020, and "Post-Covid" includes all months after March 2020. Shaded areas correspond to the 95 percent confidence intervals around point estimates. Standard errors are clustered at the loan level. Sources: Trepp and authors' calculations.

Figure A.2: Dynamic Difference-in-Differences: Trading of CMBS Exposed to Cash Flow Risks



Notes: The plot shows the time-varying effect of exposure to underlying office lease expiration (relative to 2019) on the likelihood of sales of private-label CMBS by insurers, as in (4). Shaded area corresponds to the 95 percent confidence interval around point estimates. Standard errors are clustered at the security level. Source: Trepp, NAIC, and authors' calculations.

Figure A.3: Bank Holdings of Private-label CMBS



Notes: This figure shows the total amount of private-label CMBS holdings of U.S. banks, including held-to-maturity and available-for-sale assets. Top 4 banks are J.P. Morgan Chase, Bank of America, Citigroup, and Wells Fargo. Large banks are institutions with total assets above $100 billion, medium banks are institutions with total assets between $10 billion and $100 billion, and small banks are institutions with total assets under $10 billion. We also exclude TD Bank from the plots as it shows discontinuity in holdings of private CMBS in 2018 that is not present on the aggregate series. Source: Call Reports and authors' calculations.

Table A.1: Property Types and Lease Expiration Information

Property Category	# without lease expiration	# with lease expiration	% with lease expiration
Healthcare-Nursing	530801	87	0.02
Industrial-WH	175112	69100	28.30
Lodging-Restaurants	251841	222	0.09
Mixed Use	77385	66193	46.10
Multifamily	5680699	1013	0.02
Office	80975	255201	75.91
Other	280050	14543	4.94
Retail	130208	518360	79.92

Notes: This table shows the number of observations in our CRE mortgage sample for which the lease expiration information is included, and the number of observations for which the lease expiration information is missing. Sample is from Jan/2017 to Jun/2022. Breakdown is provided by property type. Source: Trepp and authors' calculations.

Table A.2: WFH Exposure and Lease-Rollover Risk

	I_{jrt}^{D90}					
	(1)	(2)	(3)	(4)	(5)	(6)
$Post\ Expiration_{jt}$	0.0134***	0.0135***	0.0163***	0.0099**	0.0100**	0.0101**
	(0.0043)	(0.0043)	(0.0049)	(0.0040)	(0.0040)	(0.0046)
$Post\ Covid_t \times Post\ Expiration_{jt}$	0.0012	0.0011	0.0043	-0.0083*	-0.0083*	-0.0070
	(0.0048)	(0.0048)	(0.0056)	(0.0047)	(0.0047)	(0.0055)
$Post\ Covid_t \times Ind\ Office_j$	-0.0112***	-0.0111***	-0.0085**	-0.0224***	-0.0222***	-0.0309***
	(0.0029)	(0.0029)	(0.0040)	(0.0028)	(0.0028)	(0.0043)
$Post\ Expiration_{jt} \times Ind\ Office_j$	0.0209*	0.0209*	0.0168	-0.0076	-0.0075	-0.0075
	(0.0124)	(0.0124)	(0.0129)	(0.0066)	(0.0066)	(0.0068)
$Post\ Covid_t \times Post\ Expiration_{jt} \times Ind\ Office_j$	0.0055	0.0053	0.0019	0.0192**	0.0192**	0.0206**
	(0.0115)	(0.0115)	(0.0124)	(0.0078)	(0.0078)	(0.0084)
Observations	371,903	371,903	331,854	437,307	437,307	402,420
R^2	0.45443	0.45446	0.55539	0.40618	0.40627	0.49416
Month-year fixed effects	✓			✓		
Loan ID fixed effects	✓	✓	✓	✓	✓	✓
Month-year-Floating fixed effects		✓	✓		✓	✓
Month-year-City fixed effects			✓			✓

Notes: This table shows the effects of lease expiration on delinquency rates for mortgages linked to different property types, before and after COVID-19. The level of observation is loan j in city r in year-month t, which refers to the distribution month of each securitized mortgage. The sample period is Jan/2017 to Jun/2022. The dependent variable, I_{jrt}^{D90}, is a dummy variable which equals 1 if a loan is at least 90 days past due. $Post\ Covid_t$ equals 1 after March 2020, $Post\ Expiration_{jt}$ equals 1 if loan j had its main lease expiration before or in year-month t, and $Ind\ Office_j$ equals 1 if loan j is linked to an office. Columns 1-3 restrict the sample to low WFH exposure areas, defined according to the Dingel and Neiman (2020) teleworkable share. Columns 4-6 restrict the sample to high WFH exposure areas, defined according to the Dingel and Neiman (2020) teleworkable share. Standard errors clustered at the loan level in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels. Sources: Trepp and authors' calculations.

Table A.3: CMBS Characteristics and Insurer Sales

	I_{ijt}^{sold}						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$I_{jt}^{Watchlist}$	0.0167***						
	(0.0037)						
$Watchlist(\%)_{jt}$		0.0002**					
		(7.9×10^{-5})					
$Occupancy\ Rate_{jt}$			-0.0004**				
			(0.0002)				
$\Delta Occupancy_{jt}$				-0.0003**			
				(0.0002)			
$\Delta NOI(\%)_{jt}$					-5.71×10^{-5}***		
					(1.59×10^{-5})		
$Current\ DSCR_{jt}$						-0.0064*	
						(0.0035)	
$\Delta Current\ DSCR_{jt}$							-0.0230***
							(0.0045)
Observations	205,572	206,063	180,950	176,958	145,881	194,535	192,547
R^2	0.52948	0.52924	0.53677	0.53839	0.54304	0.53126	0.53200
Combined Fixed Effects	✓	✓	✓	✓	✓	✓	✓

Notes: This table shows the association between collateral-performance characteristics and insurers' likelihood of selling private-label CMBS. The level of observation is bond j held by insurer i at the end of year t. The sample period is 2017 to 2022. The dependent variable, I_{ijt}^{sold}, is a dummy which equals 1 if bond j was sold by insurer i in year t. $I_{jt}^{Watchlist}$ is a dummy which equals 1 if bond j contains any loans which are included in servicer's watchlist in year t. $Watchlist(\%)_{jt}$ corresponds to the share of bond j's deal in year t that is under servicer's watchlist. $Occupancy\ Rate_{jt}$ denotes the average occupancy rate of properties linked to bond j in year t. $\Delta Occupancy_{jt}$ is the annual change in average occupancy rate for bond j's deal in year t. $\Delta NOI(\%)_{jt}$ is the change in Net Operating Income of bond j in year t. $Current\ DSCR_{jt}$ denotes bond j's Debt Service Coverage Ratio in year t, and $\Delta Current\ DSCR_{jt}$ corresponds to the annual change in DSCR of bond j in year t. The combined fixed effects include $Year \times InsurerID$, $CUSIP \times InsurerID$, $Year \times Coupon\ Type$, $Year \times NAIC\ Designation$ and $Year \times Lagged\ Downgrade$ fixed effects. Coupon Type is the type of coupon payment for bond j (e.g., fixed rate, floating rate, interest only). Lagged downgrade is a dummy which equals 1 if the bond was downgraded (based on NAIC designation) in year $t-1$. Standard errors clustered at the security level in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels. Sources: Trepp, NAIC, and authors' calculations.

Table A.4: Exposure to Cash Flow Shocks and Interest Shortfall (Extensive Margin)

	$I_{jt}^{Interest\ Shortfall}$			
	(1)	(2)	(3)	(4)
$Treat_{jt}^{Exp\ Office}$	-0.0016*	-0.0015	-0.0039**	-0.0031**
	(0.0009)	(0.0009)	(0.0016)	(0.0015)
$Post\ Covid_t^{2021,2022} \times Treat_{jt}^{Exp\ Office}$	0.0045***	0.0027***	0.0032	0.0034**
	(0.0011)	(0.0010)	(0.0023)	(0.0015)
I_{jt}^{Office}		0.0329**	0.0330**	0.0301**
		(0.0133)	(0.0133)	(0.0133)
$Post\ Covid_t^{2021,2022} \times I_{jt}^{Office}$		0.0031**	0.0030**	0.0027**
		(0.0014)	(0.0014)	(0.0014)
I_{jt}^{Exp}			0.0039*	
			(0.0022)	
$Post\ Covid_t^{2021,2022} \times I_{jt}^{Exp}$			-0.0005	
			(0.0024)	
$I_{jt}^{Exp\ Retail}$				0.0041**
				(0.0021)
$I_{jt}^{Exp\ Other}$				-0.0012
				(0.0012)
I_{jt}^{Retail}				0.0378**
				(0.0179)
$Post\ Covid_t^{2021,2022} \times I_{jt}^{Exp\ Retail}$				0.0008
				(0.0022)
$Post\ Covid_t^{2021,2022} \times I_{jt}^{Exp\ Other}$				0.0014
				(0.0014)
$Post\ Covid_t^{2021,2022} \times I_{jt}^{Retail}$				-0.0042***
				(0.0013)
Observations	206,059	206,059	206,059	206,059
R^2	0.48667	0.48735	0.48744	0.48781
Combined Fixed Effects	✓	✓	✓	✓

Notes: This table shows the effect of exposure to underlying lease expiration and offices on bond interest shortfall. The level of observation is bond j held by insurer i at the end of year t. The sample period is 2017 to 2022. The dependent variable, $I_{jt}^{Interest\ Shortfall}$, is a dummy which equals 1 if bond j experiences interest shortfall in year t. $Post\ Covid_t^{2021,2022}$ equals 1 in 2021-2022. $Treat_{jt}^{Exp\ Office}$ is a dummy which equals 1 if bond j is exposed to office mortgages whose main lease expires within six years (excluding year t). I_{jt}^{Exp}, $I_{jt}^{Exp\ Retail}$, and $I_{jt}^{Exp\ Other}$ are dummy variables which equal 1 if bond j is exposed to mortgages whose main lease expires within six years (excluding year t), for all properties, retail only, and properties that are neither retail nor offices. I_{jt}^{Office} and I_{jt}^{Retail} are dummies which equal 1 for any exposure to offices and retail. The combined fixed effects include $Year \times InsurerID$, $CUSIP \times InsurerID$, $Year \times Coupon\ Type$, $Year \times NAIC\ Designation$ and $Year \times Lagged\ Downgrade$ fixed effects. Coupon Type is the type of coupon payment for bond j (e.g., fixed rate, floating rate, interest only). Lagged downgrade is a dummy which equals 1 if the bond was downgraded (based on NAIC designation) in year $t-1$. Standard errors clustered at the security level in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels. Sources: Trepp, NAIC, and authors' calculations.

Table A.5: Bond Characteristics - Baseline

	Treatment	Control
Office %	28.76	28.17
Retail %	28.64	31.43
Average Occupancy	90.64	92.1
Watchlisted	0.96	0.33
DSCR (Net Operating Income)	2.19	2.7
Subordination %	27.77	34.05
Non-senior	0.26	0.41
Conduit	0.95	0.28

Table A.6: CMBS Trading Difference-in-Differences—Robustness

	I_{ijt}^{sold}						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$Treat_{jt}^{Exp\ Office}$	0.0022	0.0029	0.0034	0.0044	0.0044	0.0055	0.0015
	(0.0054)	(0.0053)	(0.0053)	(0.0057)	(0.0053)	(0.0053)	(0.0059)
$I_{jt}^{Exp\ Retail}$	0.0055	0.0076	0.0088	0.0117*	0.0087	0.0114*	0.0109
	(0.0063)	(0.0063)	(0.0063)	(0.0069)	(0.0063)	(0.0063)	(0.0068)
$I_{jt}^{Exp\ Other}$	0.0073	0.0060	0.0056	0.0046	0.0070	0.0085*	0.0053
	(0.0045)	(0.0044)	(0.0044)	(0.0047)	(0.0044)	(0.0044)	(0.0047)
I_{jt}^{Retail}	-0.0178	-0.0198	-0.0073	-0.0007	-0.0140	0.0029	-0.0043
	(0.0199)	(0.0190)	(0.0206)	(0.0328)	(0.0187)	(0.0193)	(0.0349)
I_{jt}^{Office}	0.0034	0.0050	0.0059	-0.0027	0.0077	0.0203	0.0101
	(0.0193)	(0.0191)	(0.0180)	(0.0202)	(0.0174)	(0.0176)	(0.0229)
$Post\ Covid_t \times Treat_{jt}^{Exp\ Office}$	0.0170**	0.0180***	0.0158**	0.0180**	0.0171***	0.0129**	0.0154*
	(0.0072)	(0.0069)	(0.0069)	(0.0073)	(0.0065)	(0.0065)	(0.0079)
$Post\ Covid_t \times I_{jt}^{Exp\ Retail}$	0.0013	0.0121	0.0046	0.0042	0.0068	-0.0043	-0.0200
	(0.0108)	(0.0110)	(0.0105)	(0.0112)	(0.0086)	(0.0090)	(0.0134)
$Post\ Covid_t \times I_{jt}^{Exp\ Other}$	0.0034	0.0037	0.0029	0.0062	0.0031	-0.0002	0.0038
	(0.0047)	(0.0047)	(0.0047)	(0.0050)	(0.0047)	(0.0048)	(0.0051)
$Post\ Covid_t \times I_{jt}^{Retail}$	0.0100	0.0036	0.0050	0.0038	0.0011	-0.0165*	0.0015
	(0.0097)	(0.0091)	(0.0093)	(0.0098)	(0.0079)	(0.0086)	(0.0107)
$Post\ Covid_t \times I_{jt}^{Office}$	0.0151*	0.0075	0.0066	0.0073	0.0039	-0.0109	0.0029
	(0.0079)	(0.0076)	(0.0078)	(0.0080)	(0.0072)	(0.0080)	(0.0088)
$Post\ Covid_t \times Average\ Occupancy(\%)_j^{Baseline}$	-0.0003						-0.0003
	(0.0004)						(0.0005)
$Post\ Covid_t \times I_j^{Watchlist,Baseline}$		-0.0117*					-0.0166**
		(0.0069)					(0.0084)
$Post\ Covid_t \times DSCR_j^{Baseline}$			0.0005				-0.0013
			(0.0024)				(0.0029)
$Post\ Covid_t \times Subordination(\%)_j^{Baseline}$				0.0003			0.0003
				(0.0002)			(0.0002)
$Post\ Covid_t \times I_j^{Non-Senior}$					-0.0057		0.0010
					(0.0056)		(0.0061)
$Post\ Covid_t \times I_j^{Conduit}$						0.0440***	0.0573***
						(0.0096)	(0.0117)
Observations	172,996	184,071	182,052	170,507	206,063	206,063	159,007
R^2	0.52837	0.52612	0.52609	0.52851	0.52970	0.52980	0.53121
Combined Fixed Effects	✓	✓	✓	✓	✓	✓	✓

Notes: This table shows the effect of the magnitude of exposure to underlying lease expiration and offices on the likelihood of sales of private-label CMBS by insurance companies. The level of observation is bond j held by insurer i at the end of year t. The sample period is 2017 to 2022. The dependent variable, I_{ijt}^{sold}, is a dummy which equals 1 if bond j was sold by insurer i in year t. $Post\ Covid_t$ equals 1 after 2019. $Treat_{jt}^{Exp\ Office}$ is a dummy which equals 1 if bond j is exposed to office mortgages whose main lease expires within six years (excluding year t). I_{jt}^{Exp}, $I_{jt}^{Exp\ Retail}$, and $I_{jt}^{Exp\ Other}$ are dummy variables which equal 1 if bond j is exposed to mortgages whose main lease expires within six years (excluding year t), for all properties, retail only, and properties that are neither retail nor offices. I_{jt}^{Office} and I_{jt}^{Retail} are dummies which equal 1 for any exposure to offices and retail. $Average\ Occupancy(\%)_j^{Baseline}$ is the average % occupancy of properties in the collateral pool of bonds j. $I_j^{Watchlist,Baseline}$ is a dummy which equals 1 if at least one mortgage linked to bond j is flagged as under servicer watchlist in the baseline period. $DSCR_j^{Baseline}$ is the DSCR of bond j. $Subordination(\%)_j^{Baseline}$ corresponds to the subordination % of bond j. $I_j^{Non-Senior}$ is a dummy variable which equals 1 if bond j is non-senior. $I_j^{Conduit}$ is a dummy variable which equals 1 for conduit bonds. All variables correspond to averages across the pre-treatment sample, from 2017-2019. The combined fixed effects include $Year \times InsurerID$, $CUSIP \times InsurerID$, $Year \times Coupon\ Type$, $Year \times NAIC\ Designation$ and $Year \times Lagged\ Downgrade$ fixed effects. Coupon Type is the type of coupon payment for bond j (e.g., fixed rate, floating rate, interest only). Lagged downgrade is a dummy which equals 1 if the bond was downgraded (based on NAIC designation) in year $t-1$. Standard errors clustered at the security level in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels. Sources: Trepp, NAIC, and authors' calculations.

Table A.7: CMBS Trading—Exposure to Office Lease Expiration (Terciles)

	I_{ijt}^{sold}			
	(1)	(2)	(3)	(4)
$Office\ PDV\ Tercile_{jt}$ = Q2	0.0045	0.0042	-0.0022	-0.0001
	(0.0037)	(0.0038)	(0.0043)	(0.0044)
$Office\ PDV\ Tercile_{jt}$ = Q3	0.0125***	0.0118***	0.0017	0.0043
	(0.0043)	(0.0046)	(0.0058)	(0.0055)
$Post\ Covid_t \times Office\ PDV\ Tercile_{jt}$ = Q2	0.0058	0.0069	0.0107**	0.0035
	(0.0043)	(0.0046)	(0.0050)	(0.0050)
$Post\ Covid_t \times Office\ PDV\ Tercile_{jt}$ = Q3	0.0101**	0.0130**	0.0194***	0.0112*
	(0.0044)	(0.0054)	(0.0064)	(0.0059)
$Office\ Tercile_{jt}$ = Q2		-0.0017	-0.0014	-0.0011
		(0.0066)	(0.0066)	(0.0067)
$Office\ Tercile_{jt}$ = Q3		-0.0110	-0.0100	-0.0106
		(0.0088)	(0.0089)	(0.0091)
$Post\ Covid_t \times Office\ Tercile_{jt}$ = Q2		-0.0019	-0.0036	0.0006
		(0.0045)	(0.0046)	(0.0047)
$Post\ Covid_t \times Office\ Tercile_{jt}$ = Q3		-0.0044	-0.0064	0.0022
		(0.0053)	(0.0054)	(0.0059)
$All\ PDV\ Tercile_{jt}$ = Q2			0.0119***	
			(0.0040)	
$All\ PDV\ Tercile_{jt}$ = Q3			0.0127***	
			(0.0047)	
$Post\ Covid_t \times All\ PDV\ Tercile_{jt}$ = Q2			-0.0066	
			(0.0047)	
$Post\ Covid_t \times All\ PDV\ Tercile_{jt}$ = Q3			-0.0083	
			(0.0055)	
$Retail\ PDV\ Tercile_{jt}$ = Q2				0.0081
				(0.0050)
$Retail\ PDV\ Tercile_{jt}$ = Q3				0.0101*
				(0.0057)
$Other\ PDV\ Tercile_{jt}$ = Q2				0.0052
				(0.0055)
$Other\ PDV\ Tercile_{jt}$ = Q3				0.0108*
				(0.0060)
$Retail\ Tercile_{jt}$ = Q2				-0.0054
				(0.0072)
$Retail\ Tercile_{jt}$ = Q3				-0.0049
				(0.0098)
$Post\ Covid_t \times Retail\ PDV\ Tercile_{jt}$ = Q2				-0.0019
				(0.0058)
$Post\ Covid_t \times Retail\ PDV\ Tercile_{jt}$ = Q3				0.0018
				(0.0066)
$Post\ Covid_t \times Other\ PDV\ Tercile_{jt}$ = Q2				0.0150***
				(0.0055)
$Post\ Covid_t \times Other\ PDV\ Tercile_{jt}$ = Q3				0.0187***
				(0.0054)
$Post\ Covid_t \times Retail\ Tercile_{jt}$ = Q2				-0.0012
				(0.0055)
$Post\ Covid_t \times Retail\ Tercile_{jt}$ = Q3				0.0064
				(0.0065)
Observations	206,063	206,063	206,063	206,063
R^2	0.52940	0.52942	0.52948	0.52971
Combined Fixed Effects	✓	✓	✓	✓

Notes: This table shows the effect of higher exposure to lease expiration on the likelihood of sales of private-label CMBS by insurers. The unit of observation is bond j held by insurer i at year-end t (2017-2022). The dependent variable equals one if bond j is sold by insurer i in year t. $Post\ Covid_t$ equals 1 after 2019. $Office\ PDV\ Tercile_{jt}$ is a categorical variable for different terciles of the value of the discounted office lease expiration. Analogous variables are defined for retail, non-office/non-retail, and all properties. $Office\ Tercile_{jt}$ and $Retail\ Tercile_{jt}$ are categorical variables for different terciles of office and retail %, respectively. The combined fixed effects include $Year \times InsurerID$, $CUSIP \times InsurerID$, $Year \times Coupon\ Type$, $Year \times NAIC\ Designation$ and $Year \times Lagged\ Downgrade$ fixed effects. Coupon Type is the type of coupon payment for bond j (e.g., fixed rate, floating rate). Lagged downgrade is a dummy which equals 1 if the bond was downgraded (based on NAIC designation) in year $t-1$. Standard errors clustered at the security level in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels. Sources: Trepp, NAIC, and authors' calculations.

Table A.8: CMBS Portfolio Difference-in-Differences—Exposure to Lease Expiration

	$Share\ Par\ Value\ Sold_{ijt}$			
	(1)	(2)	(3)	(4)
$Treat_{jt}^{Exp\ Office}$	0.0030	0.0041	-5.8×10^{-5}	0.0018
	(0.0038)	(0.0039)	(0.0059)	(0.0049)
$Post\ Covid_t \times Treat_{jt}^{Exp\ Office}$	0.0229***	0.0201***	0.0252***	0.0129**
	(0.0040)	(0.0047)	(0.0072)	(0.0058)
I_{jt}^{Office}		0.0024	0.0014	0.0057
		(0.0161)	(0.0161)	(0.0163)
$Post\ Covid_t \times I_{jt}^{Office}$		0.0061	0.0059	0.0043
		(0.0066)	(0.0066)	(0.0066)
I_{jt}^{Exp}			0.0058	
			(0.0063)	
$Post\ Covid_t \times I_{jt}^{Exp}$			-0.0066	
			(0.0072)	
$I_{jt}^{Exp\ Retail}$				0.0034
				(0.0062)
$I_{jt}^{Exp\ Other}$				0.0054
				(0.0051)
I_{jt}^{Retail}				-0.0056
				(0.0161)
$Post\ Covid_t \times I_{jt}^{Exp\ Retail}$				0.0110
				(0.0079)
$Post\ Covid_t \times I_{jt}^{Exp\ Other}$				0.0038
				(0.0047)
$Post\ Covid_t \times I_{jt}^{Retail}$				-0.0005
				(0.0070)
Observations	204,966	204,966	204,966	204,966
R^2	0.54586	0.54586	0.54587	0.54597
Combined Fixed Effects	✓	✓	✓	✓

Notes: This table shows the effect of exposure to underlying lease expiration and offices on share of the par value of bonds sold in a given year. We estimate the same specification as in (3), but with the share of the par value of a bond sold as the dependent variable. The level of observation is bond j held by insurer i at the end of year t. The sample period is 2017 to 2022. The dependent variable, $Share\ Par\ Value\ Sold_{ijt}$, is the par value of the sales of bond j divided by the par value of the same bond in the previous year. $Post\ Covid_t$ equals 1 after 2019. $Treat_{jt}^{Exp\ Office}$ is a dummy which equals 1 if bond j is exposed to office mortgages whose main lease expires within six years (excluding year t). I_{jt}^{Exp}, $I_{jt}^{Exp\ Retail}$, and $I_{jt}^{Exp\ Other}$ are dummy variables which equal 1 if bond j is exposed to mortgages whose main lease expires within six years (excluding year t), for all properties, retail only, and properties that are neither retail nor offices. I_{jt}^{Office} and I_{jt}^{Retail} are dummies which equal 1 for any exposure to offices and retail. The combined fixed effects include $Year \times InsurerID$, $CUSIP \times InsurerID$, $Year \times Coupon\ Type$, $Year \times NAIC\ Designation$ and $Year \times Lagged\ Downgrade$ fixed effects. Coupon Type is the type of coupon payment for bond j (e.g., fixed rate, floating rate, interest only). Lagged downgrade is a dummy which equals 1 if the bond was downgraded (based on NAIC designation) in year $t-1$. Standard errors clustered at the security level in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels. Sources: Trepp, NAIC, and authors' calculations.

Table A.9: Monitoring and Insurer Portfolio Heterogeneity

	I_{ijt}^{sold}			
	(1)	(2)	(3)	(4)
$Treat_{jt}^{Exp\ Office}$	0.0099	0.0227*	0.0161**	-0.0633***
	(0.0092)	(0.0127)	(0.0067)	(0.0241)
$Post\ Covid_t \times Treat_{jt}^{Exp\ Office}$	-0.0022	-0.0027	-0.0009	0.0905***
	(0.0123)	(0.0157)	(0.0080)	(0.0299)
$Post\ Covid_t \times I_{jt}^{Exp\ Retail}$	0.0054	0.0177	0.0010	0.0362
	(0.0150)	(0.0202)	(0.0109)	(0.0362)
$Post\ Covid_t \times I_{jt}^{Exp\ Other}$	-0.0161*	0.0051	0.0056	-0.0033
	(0.0087)	(0.0111)	(0.0058)	(0.0228)
$Post\ Covid_t \times I_{jt}^{Retail}$	-0.0029	0.0049	-0.0009	0.0538
	(0.0134)	(0.0177)	(0.0106)	(0.0331)
$Post\ Covid_t \times I_{jt}^{Office}$	0.0187	0.0034	0.0116	-0.0611*
	(0.0126)	(0.0178)	(0.0099)	(0.0315)
$Post\ Covid_t \times Treat_{jt}^{Exp\ Office} \times Private\ CMBS\ Share(\%)_i$	0.0031*			
	(0.0016)			
$Post\ Covid_t \times Treat_{jt}^{Exp\ Office} \times Risky\ Share(\%)_i$		0.0006		
		(0.0005)		
$Post\ Covid_t \times Treat_{jt}^{Exp\ Office} \times Non-Senior\ Share\ CMBS(\%)_i$			0.0007***	
			(0.0002)	
$Post\ Covid_t \times Treat_{jt}^{Exp\ Office} \times Average\ Subordination\ CMBS(\%)_i$				-0.0025**
				(0.0010)
Observations	205,833	205,833	202,787	203,035
R^2	0.52984	0.52971	0.52852	0.52879
Combined Fixed Effects	✓	✓	✓	✓

Notes: This table shows the effect of exposure to underlying lease expiration and offices on the likelihood of sales of private-label CMBS by insurance companies. The level of observation is bond j held by insurer i at the end of year t. The sample period is 2017 to 2022. The dependent variable, I_{ijt}^{sold}, is a dummy which equals 1 if bond j was sold by insurer i in year t. $Post\ Covid_t$ equals 1 after 2019. $Treat_{jt}^{Exp\ Office}$ is a dummy which equals 1 if bond j is exposed to office mortgages whose main lease expires within six years (excluding year t). $Subordination(\%)_{jt}$ is the subordination percentage of bond j in year t. I_{jt}^{Exp}, $I_{jt}^{Exp\ Retail}$, and $I_{jt}^{Exp\ Other}$ are dummy variables which equal 1 if bond j is exposed to mortgages whose main lease expires within six years (excluding year t), for all properties, retail only, and properties that are neither retail nor offices. I_{jt}^{Office} and I_{jt}^{Retail} are dummies which equal 1 for any exposure to offices and retail. $Private\ CMBS\ Share(\%)_i$ denotes the % share of insurer i's fixed income assets invested in private-label CMBS. $Risky\ Share(\%)_i$ denotes the % share of insurer i's fixed income assets invested in risky assets, which are classified as assets with NAIC designation digit greater than 2. $Non-Senior\ Share\ CMBS(\%)_i$ denotes the % share of insurer i's private-label CMBS that is invested in non-senior tranches. $Average\ Subordination\ CMBS(\%)_i$ is the average subordination % of the private-label CMBS tranches held by each insurer i. The combined fixed effects include $Year \times InsurerID$, $CUSIP \times InsurerID$, $Year \times Coupon\ Type$, $Year \times NAIC\ Designation$ and $Year \times Lagged\ Downgrade$ fixed effects. Coupon Type is the type of coupon payment for bond j (e.g., fixed rate, floating rate, interest only). Lagged downgrade is a dummy which equals 1 if the bond was downgraded (based on NAIC designation) in year $t-1$. Standard errors clustered at the security level in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels. Sources: Trepp, NAIC, and authors' calculations.

Table A.10: Bond Coupons

	$Coupon_{jt}$			
	(1)	(2)	(3)	(4)
$Office_j\%$	-0.0024***	-0.0097***	-0.0069***	-0.0057***
	(0.0009)	(0.0028)	(0.0024)	(0.0021)
$Office_j\% \times Post\ Covid_t$	-0.0009	-0.0154***	-0.0157***	-0.0159***
	(0.0017)	(0.0043)	(0.0037)	(0.0033)
$NAIC\ Held_{jt}$		-0.0264	0.2471***	-0.2215**
		(0.0925)	(0.0819)	(0.1075)
$Office_j\% \times NAIC\ Held_{jt}$		0.0096***	0.0076***	0.0075***
		(0.0029)	(0.0025)	(0.0021)
$Post\ Covid_t \times NAIC\ Held_{jt}$		-0.8325***	-0.8245***	-0.8244***
		(0.1314)	(0.1152)	(0.1581)
$Post\ Covid_t \times Office_j\% \times NAIC\ Held_{jt}$		0.0190***	0.0195***	0.0142***
		(0.0042)	(0.0037)	(0.0033)
$Prime\ rating_j$			-0.7225***	-0.1233**
			(0.0349)	(0.0513)
$Main\ state\ (share\ in\ \%)_j$			-0.0027***	-0.0014
			(0.0010)	(0.0012)
$Num\ Loans\ at\ Securitization_j$			0.0003	-0.0006
			(0.0012)	(0.0015)
$Horizontal\ Risk\ Retention_j$			0.0564*	0.0661*
			(0.0323)	(0.0355)
$Retail_j\%$				-0.0179***
				(0.0036)
$Subordination\ \%_j$				-0.0181***
				(0.0016)
$Weighted\ Avg\ LTV\ at\ Securitization_j$				-0.0081**
				(0.0039)
$Weighted\ Avg\ DSCR\ at\ Securitization_j$				-0.1587***
				(0.0564)
$Conduit_j$				-0.2638**
				(0.1157)
$Post\ Covid_t \times Retail_j\%$				-0.0054
				(0.0060)
$NAIC\ Held_{jt} \times Retail_j\%$				0.0222***
				(0.0034)
$Post\ Covid_t \times NAIC\ Held_{jt} \times Retail_j\%$				0.0112*
				(0.0059)
Observations	3,200	3,200	3,200	2,449
R^2	0.56971	0.61103	0.66654	0.68199
Within R^2	0.00365	0.09932	0.21827	0.31013
Year-quarter-Maturity fixed effects	✓	✓	✓	✓
Lead Underwriter fixed effects			✓	✓

Notes: This table shows a regression of fixed-rate bond coupons of private-label CMBS on the office collateral and insurance ownership before and after COVID-19, as in $Coupon_{jt} = \alpha_{j(maturity)t} + \beta_1 Office_j\% + \beta_2 Post\ Covid_t \times Office_j\% + \beta_3 Office_j\% \times NAIC\ Held_{jt} + \beta_4 Post\ Covid_t \times Office_j\% \times NAIC\ Held_{jt} + \gamma_1 NAIC\ Held_{jt} + \gamma_2 Post\ Covid_t \times NAIC\ Held_{jt} + \Gamma X_j + \varepsilon_{jt}$. The level of observation is bond j originated in quarter t. The sample period is 2017 to 2022. The dependent variable $Coupon_{jt}$ is the coupon rate of fixed-rate bond j (in percentage points) originated in quarter t. $Post\ Covid_t$ equals 1 after 2019. $NAIC\ Held_{jt}$ equals 1 if bond j is held by any insurer at the end of the year of origination of the respective quarter t. $Office_j$ % and $Retail_j$ % are the percent shares of each bond j's corresponding deal linked to office and retail loans. $Prime\ rating_j$ equals 1 if the bond is rated at least BBB by S&P or by Fitch, or at least Baa3 by Moody's. $Main\ State\ (share\ in\ \%)_j$ is the share of the deal invested in the main state. $Num\ of\ Loans\ at\ Securitization_j$ is the number of loans in the deal at origination. $Horizontal\ Risk\ Retention_j$ and $Conduit_j$ are dummies for deals of each type, respectively. $Subordination_j\%$ is bond j's subordination percentage at origination. $Weighted\ Avg\ LTV\ at\ Securitization_j$ and $Weighted\ Avg\ DSCR\ at\ Securitization_j$ are average LTV and DSCR weighted by loan volume within each deal. Maturity denotes the bond maturity in years. Lead Underwriter is a categorical variable indicating the lead underwriter of each deal. Standard errors clustered at the security level in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels. Sources: Trepp, NAIC, and authors' calculations.

Table A.11: Small Bank Characteristics by CMBS Ownership

	CMBS Before COVID-19	CMBS Only After COVID-19	No CMBS
Total Assets (000s)	1,716,014	1,244,300	703,296
% Non-owner occ CRE loans	16.65	17.24	13.24
% Private CMBS over total assets	1.02	1.26	0
% Private CMBS	3.44	4.67	0
% Short term securities	5.63	5.50	11.12
% US Treasury	8.03	9.48	16.45
% State and Municipal Bonds	28.47	28.91	27.56
% Other Debt Securities	6.65	6.39	3.86
% Foreign Debt Securities	0.32	0.18	0.18
% Agency MBS	21.61	19.44	14.87
Tier 1 Leverage	10.75	10.88	11.51

Notes: This table shows average values for selected characteristics for three types of small banks over the four quarters of 2023. The first column includes all banks that hold private-label CMBS between 2017 and March 2020. The second column includes all banks that hold private-label CMBS only after March 2020. The last column includes all remaining banks, i.e., banks that do not hold any private-label CMBS between 2017 and 2023. Source: Call Reports and authors' calculations.

B. Data Construction

Our data come from two main sources, Trepp and NAIC, and are complemented by Call Reports data for our bank level analysis. In what follows, we document the data cleaning procedures for each of the two data sources, and show how we obtain measures of exposure to cash flow shocks at the CMBS level.

Trepp CRE mortgage data. Mortgage data is informed at the loan level with frequency determined by distribution dates (*ddate*). We use these distribution dates as our main date variables in the loan level analysis. In constructing our sample for the analysis, we exclude:

- Observations without *city* information;
- Observations with an outstanding balance lower than $ 500;
- Observations for which lease expiration is patchy, that is, when lease expiration information exists for certain months, ceases to be included, and is again included afterwards;
- Observations which have more than one broad property type associated with it in the year in our sample.

Furthermore, we use information from the variable *proptype*, informed by Trepp, to construct the broad property types which we use in our analysis. The variable *proptype* has a large number of strings indicating the use of the property serving as collateral for each mortgage. We aggregate these strings into eight different property types: *Office*, *Retail*, *Multifamily*, *Mixed Use*, *Healthcare-Nursing*, *Lodging-Restaurants*, *Industrial and Warehouses*, and the residual category *Other*. Examples of how we bin different *proptype* into our broader property type category are:

- **Office** includes *proptype* strings such as "Office", "Office/Hdqr","Office Building" and "office properties";
- **Retail** includes *proptype* strings such as "Retail", "Retail Unanchored", "Retail Anchored" and "Retail Mall";

- **Multifamily** includes *proptype* strings such as "Multi-Tenant" and "Multi-family";
- **Mixed Use** includes *proptype* strings such as "Mixed-Use", "Office/Warehouse", "Multifamily/Retail" and "Offc/Retail/Mltfmly";
- **Healthcare-Nursing** includes *proptype* strings such as "Nursing Home", "Assisted Living" and "Medical Office";
- **Lodging-Restaurants** includes *proptype* strings such as "Hospitality", "Lodging Full Service", "Restaurant" and "Hotel";
- **Industrial and Warehouses** includes *proptype* strings such as "Industrial", "Self-Storage", "Warehouse" and "Industrial/warehouse".

The full list of strings and their respective classification can be obtained upon request. Following this procedure, we obtain the loan level monthly panel summarized in Table 1.

Call Reports. We obtain bank level data at quarterly frequency from the Reports of Condition and Income (call reports), available at https://cdr.ffiec.gov/public/ManageFacsimiles.aspx. We construct our series of holdings of private CMBS by following the construction of the LM763063653.Q and LM763063693.Q variables at the bank level. Detailed instructions for the construction of these two series can be found here https://www.federalreserve.gov/apps/fof/SeriesAnalyzer.aspx?s=LM763063653&t= and in https://www.federalreserve.gov/apps/fof/SeriesAnalyzer.aspx?s=LM763063693&t=.[10]

We define *short-duration asset holdings* as the reported value of debt securities with remaining maturity of less than one year. We define *long-duration asset holdings* as the sum of reported values of government securities with over 5 years maturity (lines A553 and A554) plus the sum of mortgage pass-through securities with over 5 years maturity (lines A559 and A560).

[10] We exclude TD Bank from the analysis as its holdings of private-label CMBS suddenly drop in 2018, and no discontinuous drop is observed in either of the aggregate series. Our small bank analysis is identical as TD Bank would not be classified as a small bank.

B.1. CMBS and Insurer Level Exposure to Underlying Loan Characteristics

Since NAIC data is at annual frequency and Trepp data is at distribution date frequency (monthly), we follow an aggregation procedure to plug loan information into CMBS. Specifically, we collect deal level information corresponding to December of each year (and June for 2022, the last month in our sample from Trepp), and add this information to the bonds linked to each deal.

Specifically, let $TotAmt_{djt}$ denote the total amount outstanding of the pool of loans of deal d which is linked to bond j and $TotAmt_{djt}^{Offices}$ denote the same amount for loans linked to office properties. Then bond j's exposure to offices in year t is defined as $T_{jt}^{Office} \equiv \frac{TotAmt_{djt}^{Offices}}{TotAmt_{djt}}$. This exposure variable is used to construct dummy variables for positive exposure to offices using variables analogous to $TotAmt_{djt}^{Offices}$ that only include amount for loans with leases expiring within τ years.

To obtain insurer level exposures, we calculate a weighted average exposure at the bond level (weighted by BACV), times the size of the portfolio of private-label CMBS for each insurer.

C. Identifying Active Sales and Acquisitions

The results in Section 5 rely on measures of active asset sales and acquisitions by insurers, obtained from NAIC Schedule D, parts 3 and 4. We identify active sales using a procedure similar to Becker, Opp and Saidi (2022). First, we use the information contained in the variable *name of the purchaser* to exclude entries with keywords associated with *maturity*, *redemption*, *repayment*, *default* and *reinsurance*, for example. We also impose the requirement of strict positive or negative value in the variable *realized gain(loss) on disposal*. Finally, we further exclude observations for which maturity dates coincide with the report date.

To classify active acquisitions, we identify a series of keywords for the *vendor* variable which contain information not associated with active acquisitions. These keywords include references to *exchange*, *capitalization*, *merger* and *transfer*, for example. The full list of keywords, alongside the R code, can be obtained from the authors upon request.